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HTML, CSS, Bootstrap, Javascript and jQuery



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Chapter 1

HTML

Introduction

In this chapter, various component of HTML are discussed to design a web page. The basic structure for an HTML page is shown below.

- Entries inside the `<.../>` are known as tags. Most of the tags has an opening and closing e.g. `<head>` (opening head) and `</head>` (closing head). Some of the tags do not have closing tags e.g. `<!DOCTYPE ...>` and `<br />`. We need to write the HTML codes inside the tags.
- The comments are written between `'<!--'` and `'-->'`.
- Here Line 1 gives the details of the 'HTML version' to the web-browser. The 'html' tells it is version 5.
- The 'head' tag (Lines 3-5) contains the header related tags e.g. 'title for the page' and 'links for the css files' etc.
- The 'body' tag (7-11) contains the actual HTML code which is displayed on the web-browser. Also, we add all the JavaScript related codes just before the closing body tag (`</body>`).

```
<!DOCTYPE html> <!-- tells browser above the html version -->
<html> <!-- beginning of the html document -->
  3 <head>
  4   <!-- header related tags e.g. title, links etc. -->
  5 </head>
  6
  7 <body>
  8   <!-- actual html document here -->
  9
 10   <!-- add JavaScript files here -->
 11 </body>
12 </html>
```

First code

In below code, the message 'Hello World' is displayed on the HTML page. The Fig. 1.1 is the resultant HTML page.

- The title (Line 4) appears on the top of the browser.
- The tag `<h1>` is called 'header' tag, which has the larger size than the normal text (see the size of 'Hello World!').
- The tag `<p>` is called the 'paragraph' tag, which can be used to write the

```
<!DOCTYPE html>
<html>
```

para graphs.

(continues on next page)

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```

<head>
  <title>HTML Tutorial</title>
</head>
<body>
  <h1>Hello World! </h1>
  <p>This is the first HTML code </p>
</body>
</html>

```



Fig. 1.1: First code

Basic tags

- The [Table 1.1](#) shows the list of tags which are required for writing the basic 'HTML' codes i.e. without any style e.g. bold, italics and numbering etc.

Table 1.1: List of basic tags

Tag	Description	Example
h1, ..., h6	Header tag h1 to h6	Hi
p	paragraphs (Line changes at the end)	Hi
span	No line change after span	Hi
div	make division between contents	...
a	hyperlink	see Section 1.9
center	Move content to center	<center> Hi </center>
br	Line break (no closing tag)	 or
hr	horizontal line (no closing tag)	<hr />

- Let's see the example of each of these tags,

Note: All the new codes are added below the previous codes in the 'body' tag. Therefore only newly added codes are shown in the tutorial.

```

<h2>Heading 2 </h2>
<h6>Heading 6 </h6>

```

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```
<p> This is paragraph </p>
```

```
<span> This is span.</span>
```

```
<span> The 'br' tag is used after span to break the line </span>
<br/>
```

```
<div style="color:blue;">
```

The 'div' tag can be used for formatting the tags inside it at once using 'style' and 'classes' etc.

```
<p> This paragraph is inside the 'div' tag </p>
```

```
<span> This span is inside the 'div' tag </span>
```

```
<br/>
```

```
</div>
```

```
<center>
```

```
<h3> Heading 3 is centered</h3>
```

```
<p><span> Centered span inside the paragraph.</span><p>
```

```
</center>
```

Two horizontal line is drawn using two 'hr' tag.

```
<hr />
```

```
<hr>
```

<pre> 'pre' tag preserve the formatting (good for writing codes)

```
# Python
code x = 2
y = 3
print(x+y)
```

Fig. 1.2 is the output of above code. Read the text to understand each tag,

Heading 2

Heading 6

This is paragraph

This is span. The 'br' tag is used after span to break the line

The 'div' tag can be used for formatting the tags inside it at once using 'style' and 'classes' etc.

This paragraph is inside the 'div' tag

This span is inside the 'div' tag

Heading 3 is centered

Centered span inside the paragraph.

Two horizontal line is drawn using two 'hr' tag.

'pre' tag preserve the formatting (good for writing codes)

```
# Python code
x = 2
y = 3
print(x+y)
```

Fig. 1.2: Basic tags : Attribute 'style' is used in 'div' tag

Attributes

In [Fig. 1.2](#), we saw an example of an attribute (i.e. `style`) which changed the color of all the elements to 'blue' inside the `'div'` tag.

Attribute 'name' and 'value'

- Attribute is defined inside the opening part of a 'tag'. For example, in the below code, the attribute `'style'` is defined inside the `'div'` tag.

```
<div style="color:blue;">
</div>
```

- An attribute has two parts i.e. **'name'** and **'value'**. For example, in the above code, **name** and **value** of the attribute are **'style'** and **'blue'** respectively.

Core attributes

Below are the three core attributes which are used frequently in web design.

- id**: The 'id' is the **unique** name which can be given to any tag. This is very useful in distinguishing the element with other elements.

```
<p id="para1"> This is paragraph with id 'para1' </p>
<p id="para2"> This is paragraph with id 'para2' </p>

<p class="c_blue"> This is paragraph with class 'blue' </p>
<span class="c_blue"> This is span with class 'blue' </span>
```

- class**: The attribute 'class' can be used with multiple tags. This is very useful in making groups in HTML design.
- style**: We already see the example of style attribute, which can be used to change the formatting of the text in HTML design. We can specify

```
<p style="font-weight:bold;color:red;">Style attribute is used to bold and color</p>
```

various styles which are discussed in [Chapter 2](#).

Note: Above three attributes are used with 'CSS (cascading style sheet)' and JavaScript/jQuery, which are the very handy tools to enhance the look and functionalities of the web-page respectively. The CSS is discussed in [Chapter 2](#), whereas JavaScript and jQuery are discussed in [Chapter 4](#) and [Chapter 5](#) respectively.

- Also we can define multiple attributes for one tag as shown below,

```
<p class="my_class" id="para_with_class" style="color:green"> Multiple attributes </p>
```

- The other useful attributes are listed in [Table 1.2](#)

Table 1.2: List of attributes

| Name | Values | Description |
|-------|--------------------------|---|
| id | user defined names | <p id=' p_l' > Hi </p> |
| class | user defined names | <p class=' p_class' > Hi </p> |
| style | CSS styles | <p style=" color:red;font-weight:bold;" > Hi </p> |
| align | left, right, center | horizontal |
| width | numeric value or % value | width of images |

Tables

In this section, we will learn to draw tables along with some attributes which are discussed in [Table 1.2](#). [Table 1.3](#) shows the list of tags available to create the table, which are used in [Listing 1.1](#).

Table 1.3: Tags and attributes for creating tables

| Tag | Description |
|-------|------------------------------|
| table | beginning and end of |
| tr | table row of table |
| th | header |
| td | cell data |
| td | .. |
| tr | number of rows to merge |
| td | number of columns to |
| tr | merge width of border |
| td | width of white space between |
| td | two border width of |
| td | white space within a border |
| td | background color |
| td | color of border |
| td | width of table (numeric |
| td | or %) height of table |
| td | (numeric) caption for |

Some of the attributes of [Table 1.3](#) are used in below example,

Listing 1.1: Table with border and color

```

<!-- border-color, width and height -->
<table border="1" bordercolor="black" width="450"
height="100">
<caption>Table 1 : Various tags of table</caption>
  5 <tr bgcolor="red" > <!-- row -->
  6   <th>Column 1</th> <!-- header -->
  7   <th>Column 2</th>
  8   <th>Column 3</th>
  9 </tr>
 10
 11 <tr bgcolor="cyan"> <!-- background color -->
 12   <td>Data 1</td> <!-- data -->
 13   <td>Data 2</td>
 14   <td>Data 3</td>
 15 </tr>
 16
 17 <tr bgcolor="yellow"> <!-- row -->
 18   <td colspan="2">New Data 1</td> <!-- column span -
 19   -->
 20   <td>New Data 2</td> <!-- data -->
 21
 22
 23
 24   %
 25 <!-- width in --> %
 26 <table border="1" bordercolor="black" width="80 "
 27 height="100">
 28 <caption> Table 2 : Width is 80 </caption>
 29 <tr bgcolor="red" >
 30   <th>Column 1</th>
 31   <th>Column 2</th>
 32   <th>Column 3</th>

```

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```
|
|  |

```

· Fig. 1.3 is the output of above code,

| Table 1 : Various tags of table | | |
|---------------------------------|----------|------------|
| Column 1 | Column 2 | Column 3 |
| Data 1 | Data 2 | Data 3 |
| New Data 1 | | New Data 2 |

| Table 2 : Width is 80% | | |
|------------------------|----------|----------|
| Column 1 | Column 2 | Column 3 |
| Data 1 | Data 2 | Data 3 |

Fig. 1.3: Table generated by Table 1.3

Text formatting

In this section, we will see some of the text formatting options (see Table 1.4) e.g. bold, italic, subscript and strike etc.

Table 1.4: Text formatting

| Tag | Description |
|--------|-------------|
| b | bold |
| i | italic |
| u, ins | underline |
| del | strike |
| sup | superscript |
| sub | subscript |

· Below are the some of the examples of text formatting, whose results

```

<!-- Text formatting -->
<p>This is <b>bold</b> text</p>
<p>This is <del>strided</del> text</p>
<p>This is <sub>subscript</sub> text</p>

```

are shown in Fig. 1.4,

Images

Image tag has two important attributes i.e. 'src' and 'alt' as described below,

· **src**: tells the location of 'image' file e.g. in Line 2 the image 'logo.jpg' will be searched inside the folder 'img'.

This is **bold** text

This is ~~striked~~ text

This is subscript text

Fig. 1.4: Text formatting

- **alt** : is the 'alternate text' which is displayed if image is not found. For example, in Line 6, the name of the image is incorrectly written i.e. 'logo a' (instead of 'logo'), therefore the value of 'alt' i.e. 'Missing Logo.jpg' will be displayed as shown in Fig. 1.5.

```
<!-- Images -->  
  
3  
<br /> <br />  
5  

```



Fig. 1.5: Images

Note: We can use other attributes as well e.g. **height**, **align** and **border** etc.

Lists

There are three type of lists in HTML,

- Unordered list: bullet are used in it (see Lines 2 and 9)
- Ordered list: numbers are used in it (see Lines 15, 22 and 28)
- Definition list: This can be used for writing definitions in HTML (see Line 35)

```

<!-- Lists -->
<!-- unordered list -->
<ul> Unordered
  4   List
  5 <li>Pen</li>
  6 <li>Pencil</li>
  7 <li>Eraser</li>
</ul>
<ul type="circle"> Change bullets : 'square', 'circle' or 'disc'
 10 <li>Pen</li>
 11 <li>Pencil</li>
 12 <li>Eraser</li>
</ul>
<!-- ordered list -->
<ol> Ordered
 17   List
 18
 19 <li>Pen</li>
 20 <li>Pencil</li>
 21 <li>Eraser</li>
</ol>
<ol type="i"> Change style : 'i', 'I', 'l', 'a' or 'A'
 24 <li>Pen</li>
 25 <li>Pencil</li>
 26 <li>Eraser</li>
</ol>
<ol type="i" start="5"> Start from 'v'
 29 <li>Pen</li>
 30 <li>Pencil</li>
 31 <li>Eraser</li>
</ol>
<!-- Definition list -->
<dl>
  34 <dt> <h4>HTML Definition List</h4> </dt>
  35 <dd> HTML is a sy</dd>
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```

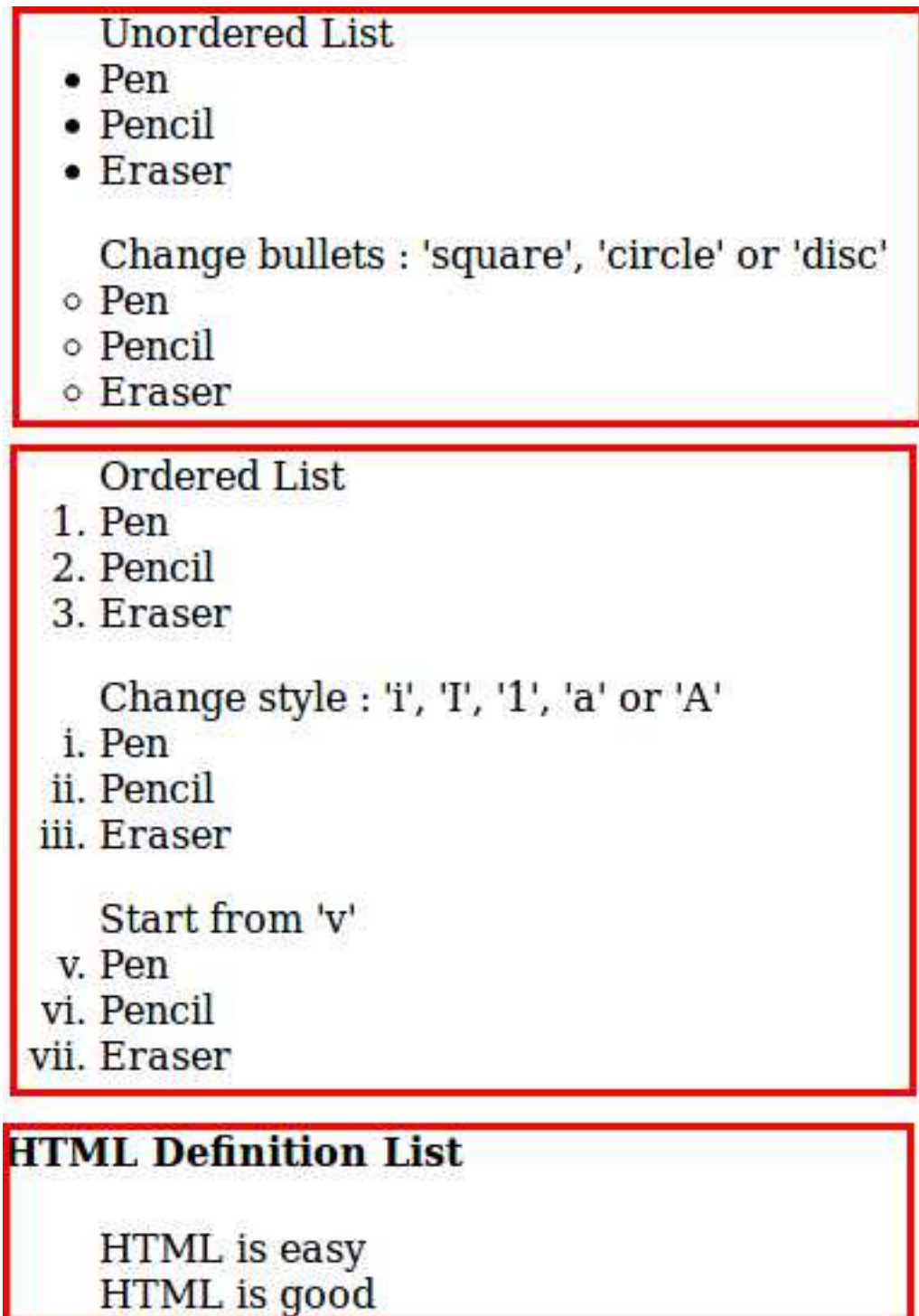


Fig. 1.6: Lists

(continued from previous page)

```
<p><a href="http://pythondsp.readthedocs.io/pdf">Download PDF, DOC or Zip
```

```
<p><a href="mailto:pythondsp@gmail.com">Email me</a></p>
```

```
<p><a href="mailto:pythondsp@gmail.com?subject=Feedback&body=Your feedback here">Feedback
```

Go to paragraph with [id='para1'](#)

[PythonDSP](#)

[JavaScript Tutorial](#) in same window.

[JavaScript Tutorial](#) in new Window.

[Download PDF, DOC or Zip Files](#)

[Email me](#)

[Feedback email](#)

Note: We can change the color of the links using 'alink (active link)', 'link' and 'vlink (visited link)', by defining these attributes in the 'body tag' as shown below,

```
<body alink="green" link="blue" vlink="red">
```

Forms

Forms can have different types of controls to collect the input-data from users, which are listed below and shown in [Table 1.5](#),

- Text input
- Text area
- Radio button
- Checkbox
- Select box
- File select
- Buttons
- Submit and reset buttons
- Hidden input

Table 1.5: List of control inputs and their attributes

| Control | Attributes | Values | Description |
|------------------------|------------------------------|--------------------------|------------------------------------|
| Input : text | type | text, password | |
| | value | user-defined | initial |
| | value in the area | name | user-defined |
| | name | send to server | |
| | size | numeric value | width of |
| | the text area | | |
| | maxlength | | numeric value |
| radio | maximum number of characters | Input : | |
| | type | radio | |
| | name | user-defined | name send to server |
| | value | user-defined value | value of |
| | the button if selected | checked | check |
| Input : check box | the button by default | | |
| | type | checkbox | |
| | name | user-defined | name send to server |
| | value | user-defined value | value of |
| Input : button | the box if selected | checked | |
| | | check the box by default | |
| | type | button | trigger client side script |
| | | submit | submit the form and run 'action' |
| | | reset | reset form |
| | | image | create |
| | image button method | get, post | get or post method |
| Input : hidden be used | action | user-defined | action to perform on submit |
| | type | hidden | will not display on html, but can |
| | | | for sending |
| | information to server | | |
| Text area | Selection box | name | user-defined |
| | name | send to server | |
| | size | numeric value | enable scroll |
| | (default dropdown) | multiple | numeric value |
| | | select multiple items | |
| | value | user-defined value | value of |
| | the item if selected | selected | |
| Text area | | select item by default | |
| | rows, cols | numeric value | number of |
| | rows and cols | name | user-defined |
| | | name send to server | |

- Below are the example of the control inputs described in [Table 1.5](#)
(continues on next page)

```

<!-- Forms -->
<form>
  <h4>Text input </h4>
  Name : <input type="text" name="user_name" size="4" value="e.g. meher21"
  maxlength="10"><br> Password : <input type="password" name="user_pass" ><br>

  <h4> Radio button: name should be
  same </h4>
  <input type="radio" name="r_gender"> Male
  <input type="radio" name="r_gender"> Female
  <input type="radio" name="r_gender" checked>
  Infant

  <h4> Check box : name should be
  different </h4>
  <input type="checkbox" name="c_male" checked>
  Male
  <input type="checkbox" name="c_female">
  Female
  <input type="checkbox" name="c_infant"> Infant

  <h4> Select box : drop-down </h4>
  <select name="s_box">
    <option value="s_male">Male </option>

```

(continued from previous page)

```

<select name="s_box" size="4" multiple>
  <option value="s_male" selected>Male</option>
  <option value="s_female" selected>Female</option>
  <option value="s_infant">Infant 1</option>
  <option value="s_infant" selected>Infant 2</option>
  <option value="s_infant">Infant 3</option>
  <option value="s_infant">Infant 4</option>
</select>

<h4> Text area</h4>
<textarea rows="10" cols="80"
name="txt_area">Initial Text x = 2
y = 3
</textarea> <!-- formatting work as pre -->

</form>

```

Fig. 1.7 is the output of above code,

- Below is the code which shows the working of various buttons. Note that **method** and **action** are defined in this form, which will be triggered on 'submit'

```

<form method="get|post" action="jquery.html">
  <h4> Buttons and Hidden</h4>

  Name : <input type="text" name="user_name" size="4" value="Meher"
  maxlength="16"><br> Password : <input type="password" name="user_pass"
  ><br>

  <input type="button" onclick="alert('Hello')" name="b_alert" value="Say Hello"/><br>
  <input type="submit" name="b_submit" value="Go to jQuery"/>
  <input type="reset" name="b_reset" value="Reset"/><br>

  <input type="hidden" name="h_data" value="html_tutorial">

```

button. Lastly, 'hidden' option is used in this example.

Fig. 1.8 is the output of above code,

Text inputName : Password : **Radio button: name should be same**☐ Male ☐ Female ☒ Infant**Check box : name should be different**☒ Male ☐ Female ☐ Infant**Select box : drop-down****Select box : scroll**

Text area

```
Initial Text
x = 2
y = 3
```

Fig. 1.7: Various control inputs for creating form

Buttons and Hidden

Name :

Password :

Fig. 1.8: V a r i o u s b u t t o n s a n d h i d d e n - i n p u t i n t h e f o r m

Chapter 2

Cascading Style Sheets (CSS)

Introduction

CSS is used to enhance the look of the webpage. In [Section 1.4.2](#), we saw the attribute 'style', which is used for changing the color of the text. Let's rewrite the example of 'style' as shown in next section.

Inline CSS

- Below code is an example of 'inline CSS', where the styles are defined inside the individual tags.

```
<!-- css.html -->
2
<!DOCTYPE html>
<html>
<head>
6 <title>CSS Tutorial</title>
7
</head>
8
9 <body>
10
11 <h3 style="color:blue">Heading 1 </h3>
12 <h3 style="color:blue">Heading 3 </h3>
13 <h3 style="color:blue">Heading 3 </h3>
14
15 </body>
16 </html>
```

In the above code, we have three 'headings' with font-color as 'blue'. Suppose, we want to change the color to red, then we must go to individual 'h3' tag and then change the color. This is easy in this case, but if we have 100 headings in 5 different 'html' files, then this process is not very handy. In such cases, CSS can be quite useful as shown in next section.

Embedded CSS

In the below code, the style is embedded inside the 'style' tag as shown in Lines 8-17. Here, we have defined two classes i.e. 'h3_blue (Lines 21-23)' and 'h3_red (Lines 26-28)'. Then, the **selectors** at Lines 9 and 13 targets the class 'h3_blue' & 'h3_red', and change the color to blue and red respectively. In this chapter, we will discuss the selectors (e.g. h3.h3_blue) in more details.

Note:

- In CSS, the comments are written between /* and */.

- CSS has three parts,
 - **Selectors** e.g. p, h3, h3_blue
 - **Properties** e.g. color
 - **Values** of properties e.g. red

```

<!-- css.html -->
2
<!DOCTYPE html>
<html>
<head>
6 <title>CSS Tutorial</title>
7
  <style type="text/css">
    h3.h3_blue{ /*change color to
      blue*/
      color: blue;
    }

    h3.h3_red{ /*change color to
      red*/
      color: red;
    }
19 </style>
20 >
21
</head>
23 <body>
24
25
26
  <h3 class="h3_red"> Heading 1
28 </h3>
29 <h3 class="h3_red"> Heading 3
30 </h3>
31 <h3 class="h3_red"> Heading 3
  </h3>

```

- Below is the output of above code,

External CSS

We can write the 'CSS' code in different file and then import the file into 'html' document as shown in this section. In this way, we can manage the files easily.

- The 'CSS' code is saved in the file 'my_css.css' which is saved inside the folder

```

/* asset/css/my_css.css */

h3.h3_blue{
  color: blue;
}

h3.h3_red{
  color: red;
}

```

'asset/css' .

- Next, we need to import the CSS file into the 'html' file as shown in Line 7.

```

<!-- css.html -->
2

```

(continues on next page)

Heading 1**Heading 3****Heading 3****Heading 1****Heading 3****Heading 3**

Fig. 2.1: Embedded CSS

(continued from previous page)

```

<!DOCTYPE html>
<html>
<head>
  6 <title>CSS Tutorial</title>
  <link rel="stylesheet" type="text/css" href="asset/css/my_css.css">
</head>
<body>
  10
  11 <h3 class="h3_blue">Heading 1 </h3>
  12 <h3 class="h3_blue">Heading 3 </h3>
  13 <h3 class="h3_blue">Heading 3 </h3>
  14
  15
  16 <h3 class="h3_red">Heading 1
  17 </h3>
  18 <h3 class="h3_red">Heading 3
  19 </h3>
  20 <h3 class="h3_red">Heading 3
  21 </h3>

```

Basic CSS Selectors

There are three types of selectors in CSS,

- **Element**: can be selected using its name e.g. 'p', 'div' and 'h1' etc.
- **Class**: can be selected using 'className' operator e.g. '.h3_blue'.
- **ID**: can be selected using '#idName' e.g. '#my_para'.

We will use following HTML for understanding the selectors,

```

1 <!DOCTYPE html -->
2
3 <html>
4 <head>
5   6 <title>CSS Selectors</title>
7   <link rel="stylesheet" type="text/css" href="asset/css/my_css.css">
8 </head>
9 <body>
10  <h3>CSS Selectors</h3>
11
12  <p class='c_head'> Paragraph with class 'c_head' </p>
13  <p id='i_head'> Paragraph with id 'i_head' </p>
14
15 </body>
16 </html>

```

- Below code shows the example of different selectors, and the output is shown in Fig. 2.2

```

1 /*
2  asset/css/my_css.css */
3
4 /* element
5  selection*/
6 h3{
7   . . .
8 }
9
10 /*class selection*/
11 .c_head{
12   font-family:
13   cursive; color:
14   orange;
15 }
16
17 /*id selection*/
18 #i_head{
19   font-variant: small-
20   caps; color: red;
21 }

```

CSS Selectors

Paragraph with class 'c_head'

PARAGRAPH WITH ID 'I_HEAD'

Fig. 2.2: Selectors : element, class and id

Hierarchy

In previous section, we saw the example of selectors. In this section, we will understand the hierarchy of the styling-operations.

Important: Below is the priority level for the CSS,

- Priority level :
 - ID (highest priority)
 - Class
 - Element
- If two CSS has same priority, then CSS rule at the last will be applicable.

- Below is the html code with following tags,
 - 'p' tag
 - 'p' tag with class 'c_head'
 - 'p' tag with class 'c_head' and id 'i_head'

```

<!-- css.html -->
2
<!DOCTYPE html>
<html>
<head>
6 <title>CSS Selectors</title>
7 <link rel="stylesheet" type="text/css" href="asset/css/my_css.css">
</head>
<body>
10 <p>Paragraph</p>
11
12 <p class='c_head'> Paragraph with class 'c_head' </p>
13
14 <p class='c_head' id='i_head'> Paragraph with class 'c_head' and id 'i_head' </p>
15
</body>
16
</html>

```

Below is the CSS code. Let's understand the formatting of all three 'p' tags individually. The results are shown in [Fig. 2.3](#).

- **'p' tag at Line 13 of html** : Since, 'id' has highest priority, therefore CSS rule for '#i_head' (Line 12) will not be overridden by Line 24; hence the color is **red**. Line 13 has 'p' tag, therefore 'font-variant' rule will be applied by Line 17. Also, this tag has class 'c_head', therefore 'font' will be set to 'cursive'. Hence, the line is **'all-caps with font-cursive in red color'**.
- **'p' tag at Line 12 of html** : Similarly, the 'head' tag has higher priority than 'element' therefore color of this line is **orange** and font-family is 'cursive'. Also, Line 17 will make it **all caps**.
- **'p' tag at Line 10 of html** : Color defined at Line 18 will be overridden by Line 24; hence the color will be **blue**. Also, Line 17 will make it **all caps**.

```

/* asset/css/my_css.css */
2
/* class selection */
3 .c_head{
4   font-family: cursive;
5   color: orange;    /* override the blue
6   color*/
7
8
9
10 /* id selection */
11 #i_head{
12   color: red;
13 }
14
15 /* element selection */
16 p{
17   font-variant: small-
18   caps; color: blue;
19 }
20
21

```

(continues on next page)

(continued from previous page)

```

/*element selection*/
p {
  color: green;
}

```

PARAGRAPH

PARAGRAPH WITH CLASS 'C_HEAD'

PARAGRAPH WITH CLASS 'C_HEAD' AND ID 'I_HEAD'

Fig. 2.3: Priority level for CSS rule

More selectors

Table 2.1 shows the combinations of selectors to target the various elements of the HTML. Also, some of the example of 'Attribute selector' is shown in this section.

Table 2.1: List of selectors

Selectors	Description
h1, p, span etc.	element selector
.className	class selector
#idName	id selector
*	Universal selector (selects everything)
h1.className	select h1 with class 'className'
h1#idName	select h1 with id 'idName'
p span	descendant selector (select span which is inside p)
p > span	child selector ('span' which is direct descendant of 'p')
h1, h2, p	group selection (select h1, h2 and p)
span[my_id]	select 'span' with attribute 'my_id'
span[my_id=m_span]	select 'span' with attribute 'my_id=m_span'

Attribute selector

- Add below code at the end of the html file. In these lines 'custom

```

<!-- css.html -->

<span my_id='m_span'> Span with a ttribute 'my_id' with value 'm_span' </span>
<br>
<span my_id='m_span2'> Span with a ttribute 'my_id' with value 'm_span2' </span>

```

attributes' are added (i.e. my_id).

- These custom attributes can be selected as below,

```

/* attribute selection */
span[my_id] { /* select 'span' with attribute 'my_id' */
  color: green;
  font-weight:

```

(continues on next page)

(continued from previous page)

```

}

span[my_id=m_span] { /* select 'span' with attribute 'my_id =
m_span' */
    color: red;

```

More properties

Table 2.2 shows the some more important properties which can be used in CSS,

Table 2.2: More CSS properties

Property	Syntax	Description/possible values
size	20%	size = 20%
	20px	20 pixel
	2em	2*font-size
	2mm, 2cm, 2in	2 mm, cm
	and inch color	names
	e.g. red, blue, green	hex code
	(#RRGGBB or #RGB)	#FFF000 or #FFF
	rgb(num, num, num)	rgb(0, 0, 255) or rgb(20%,
	10%, 70%) link	a:link a:link {color:
	red}	
Font	a:h	
	ov	
	er	
	a:vis	
	ited	
	a:a	
	ctiv	
	e	
	font-family	serif, cursive
	font-style	normal, italic, oblique
Text	font-variant	normal, small-caps
	font-weight	normal, bold, bolder, lighter, 100-
	900	
	font-size	10px, small, medium, large etc.
	color	red, #FFF
	letter-spacing	10px
	word-spacing	10 px
	text-align	right, left, center
	text-decoration	underline, overline, line-through,
	none	
5px 8px red; Image	text-transform	capitalize, uppercase,
	lowercase, none	
	white-space	pre, normal, nowrap
	text-shadow	text-shadow: 5px
	'1px', or '1px solid blue'	border
	height, width	100px, 20%
	border-style	solid, dashed, dotted,
	double, none etc. border-top-style	
	border-	
	bottom-	
	style	

border-left-style
border-right-style
border-width 4px, 4pt
border-bottom-width similarly use 'top', 'left', 'right'
border (shortcut) 1px
solid blue' Margin
margin, margin-left etc.
Padding padding (top, bottom, left, right) '10px 10px 2px 2px' or '10px 2px padding-right, padding-left etc.

Chapter 3

Bootstrap

Introduction

One of the problem with basic HTML design is that the webpage may look different in different browser or device (e.g. mobile, tablet and laptop). Therefore, we may need to modify the code according to browser or device. The problem can be easily resolved by using Bootstrap.

Bootstrap is a framework which uses HTML, CSS and JavaScript for the web design. It is supported by all the major browsers e.g. Firefox, Opera and Chrome etc. Further, Bootstrap includes several predefined classes for easy layouts e.g. dropdown buttons, navigation bar and alerts etc. Lastly, it is responsive in nature i.e. the layout changes automatically according to the device e.g. mobile or laptop etc.

Setup

Bootstrap needs at least 3 files for its operation which can be [downloaded from the Bootstrap website](#).

- **bootstrap.css (Line 7)** : This file contains various CSS for bootstrap.
- **bootstrap.js (Line 16)** : This file contains various JavaScript functionalities e.g. dropdown and alerts etc.
- **jQuery.js (Line 17)** : This file is the jQuery library which can be downloaded from the 'jQuery' website. It is required for proper working of 'bootstrap.js'.

Download and include files

These files are downloaded and saved inside the 'asset' folder. Next, we need to include these files in the HTML document as below,

```

<!DOCTYPE html>
<html>
<head>
  4 <title>Bootstrap Tutorial</title>
  5
  6 <!-- CSS -->
  7 <link href="asset/css/bootstrap.min.css" rel="stylesheet">
  8 <!-- Add Custom CSS below -->
  9
  10 </head>
  11 <body>
  12
  13
  14 <!-- JavaScript -->
  15 <!-- put jQuery.js before bootstrap.min.js; and then add

```


(continues on next page)

(continued from previous page)

```
<script src="asset/js/jquery-3.3.1.min.js"></script>
<script src="asset/js/bootstrap.min.js"></script>
</body>
</html>
```

Add CDN

Another way to include the files is CDN. In this method, we need not to download the files, but provide the links to these files, as shown in Lines 8, 17 and 19 of below code. Note that, in this case the code will not work in offline mode.

```
<!DOCTYPE html>
<html>
<head>
  4 <title>Bootstrap Tutorial</title>
  5
  6 <!-- CSS -->
  7 <!-- Latest compiled and minified CSS -->
    <link rel="stylesheet" href="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/css/bootstrap.min.css"
    integrity="sha384-BVYiiSIFeK1dGmJRAKycuHAHRg32OmUcww7on3RYdg4Va+PmSTsz/K68vbdEjh4u" crossorigin=
    "anonymous">
    <!-- Add Custom CSS below -->
  10
  11 </head>
  12 <body>
  13
  14
  15
  16 <!-- JavaScript -->
  17 <!-- jQuery -->
    <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.3.1/jquery.min.js"></script>
  19 <!-- Latest compiled and minified JavaScript -->
    <script src="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/js/bootstrap.min.js"
    integrity="sha384-
    Pet5IQib027qvyjSMfHjOMaLkfuWVxZxUPnCJA7l2mCWNIpG9mGCD8wGNlcpPD7Txa" crossorigin="anonymous"></script>
  </body>
```

Note: In this tutorial, we have used the first method i.e. download the files and include in the html document.

Check setup

Let's write our first code to check the correct setup of the Bootstrap.

Note: We need not to memorize the code for creating the 'dropdown' button. All the templates are available on the [Bootstrap website](#). Copy and paste the code from there and then modify the code according to need as shown in this tutorial.

```
<!DOCTYPE html>
<html>
<head>
  4 <title>Bootstrap Tutorial</title>
  5
  6 <!-- CSS -->
  7 <link href="asset/css/bootstrap.min.css" rel="stylesheet">
  8 <!-- Add Custom CSS below -->
```

(continues on next page)

(continued from previous page)

```

9
</head>
<body>
12
    <div class="dropdown">
        <button class="btn btn-success dropdown-toggle" type="button" id="dropdownMenu1" data-
            toggle=
            "dropdown" aria-haspopup="true" aria-expanded="true">
            Dropdown
            <span class="caret"></span>
        </button>
        <ul class="dropdown-menu" aria-labelledby="dropdownMenu1">
            <li><a href="#">Action</a></li>
            <li><a href="#">Another action</a></li>
            <li><a href="#">Something else here</a></li>
            <li role="separator" class="divider"></li>
            <li><a href="#">Separated link</a></li>
        </ul>
27 </div>
28
29 <!-- Javascript -->
30 <!-- put jQuery.js before bootstrap.min.js; and then add custom jQuery -->
31 <script src="asset/js/jquery-3.3.1.min.js"></script>
32 <script src="asset/js/bootstrap.min.js"></script>
</body>

```

The resultant webpage is shown in Fig. 3.1,



Fig. 3.1: Click on the 'Dropdown' button

Grid system

Bootstrap divides the **each row** into **12 columns**. Then following commands can be used to specify the width the columns

- **col-lg-4** : It will select 4 columns. Choose any number between 1-12. The 'lg' stand for **large screen** (e.g. large computer screen).
- **col-md-5** : 'md' = medium screen
- **col-sm-3** : 'sm' = small screen
- **col-xs-3** : 'xs' = extra small screen
- **col-lg-offset-4** : skip first 4 columns. Similarly use 'md', 'sm' and 'xs' with offset

Example

Below is an example of grid system. Read the content of the Lines 13-16 to understand it. The resultant webpage is shown in [Fig. 3.2](#).

Note:

- For easy visualization, in the below code the CSS code (Lines 9-21) is used to fill the columns with colors and border.
- The columns (Lines 28-30) should be defined inside the class 'row' (Line 27).
- Also, sum of the widths of individual columns should not be greater than 12.
- Lastly, if we use 'col-md-4' without defining the 'lg', 'sm' and 'xs', then 'md' rule will be applicable to higher size screen 'lg', but **not** on the lower size screen 'sm' and 'xs'. Similarly, if we use 'sm' without defining the 'lg', 'md' and 'xs', then rule

will be applicable to higher size screen i.e. 'lg' and 'md' but not on 'xs'

```
<!DOCTYPE html>
<html>
<head>
  4 <title>Bootstrap Tutorial</title>
  5
  6 <!-- CSS -->
  7 <link href="asset/css/bootstrap.min.css" rel="stylesheet">
  8 <!-- Add Custom CSS below -->
  9 <style type="text/css">
    .col-xs-1, .col-xs-2, .col-xs-3, .col-xs-4, .col-xs-5, .col-xs-6, .col-xs-7, .col-
    xs-8,
    .col-sm-1, .col-sm-2, .col-sm-3, .col-sm-4, .col-sm-5, .col-sm-6, .col-sm-7, .col-
    sm-8,
    .col-md-1, .col-md-2, .col-md-3, .col-md-4, .col-md-5, .col-md-6, .col-md-7, .col-
    md-8,
    .col-lg-1, .col-lg-2, .col-lg-3, .col-lg-4, .col-lg-5, .col-lg-6, .col-lg-7, .col-
    lg-8 { background-color: green;
    color: white;
    font-weight: bold;
    border: 1px solid red;
    height: 3em; /*height of the box*/
    text-align: center; /*vertical center the text*/
  23   }
  24 }
  25 </style>
  26 >
  27
</head>
<body>
  28 <div class="row">
  29
  30 <!-- Javascript -->
  31 <!-- put jquery.js before bootstrap.min.js; and then add
  32 custom jquery -->
  33 <script src="asset/js/jquery-3.3.1.min.js"></script>
  34 <script src="asset/js/bootstrap.min.js"></script>
  35 </body>
  36
  37 col-md-2, col-xs-4
  38 col-md-6, col-xs-4
  39 col-md-4, col-xs-4
```

Fig. 3.2: Different grid-size (i.e. 2, 6, 4) for 'medium' and 'large' screen



Fig. 3.3: Equal grid-size (i.e. 4) for ‘extra small’ and ‘small’ screen

Nested columns

We can further divide a column into small columns by defining a class ‘row (Line 30)’ inside the ‘column’ (Line 29), as shown in Lines 29-34.

```
<!DOCTYPE html>
<html>
<head>
  4 <title>Bootstrap Tutorial</title>
  5
  6 <!-- CSS -->
  7 <link href="asset/css/bootstrap.min.css" rel="stylesheet">
  8 <!-- Add Custom CSS below -->
  9 <style type="text/css">
 10   .col-xs-1, .col-xs-2, .col-xs-3, .col-xs-4, .col-xs-5, .col-xs-6, .col-xs-7, .col-
 11   xs-8,
 12   .col-sm-1, .col-sm-2, .col-sm-3, .col-sm-4, .col-sm-5, .col-sm-6, .col-sm-7, .col-
 13   sm-8,
 14   .col-md-1, .col-md-2, .col-md-3, .col-md-4, .col-md-5, .col-md-6, .col-md-7, .col-
 15   md-8,
 16   .col-lg-1, .col-lg-2, .col-lg-3, .col-lg-4, .col-lg-5, .col-lg-6, .col-lg-7, .col-
 17   lg-8 { background-color: green;
 18   color: white;
 19   font-weight: bold;
 20   border: 1px solid red;
 21   height: 3em; /*height of the box*/
 22   text-align: center; /*vertical center the text*/
 23   }
 24 </style>
 25 >
 26
 27 </head>
 28
 29 <body>
 30   <div class="row">
 31     <div class="col-md-2 col-xs-4">col-md-2, col-xs-4</div>
 32     <div class="col-md-6 col-xs-4">
 33       <div class="row">
 34         <div class="col-xs-6">col-xs-6</div>
 35         <div class="col-xs-6">col-xs-6</div>
 36       </div>
 37     </div>
 38   </div>
 39
 40   <!-- JavaScript -->
 41   <!-- put jQuery.js before bootstrap.min.js; and then add
 42   custom jQuery -->
 43   <script src="asset/js/jquery-3.3.1.min.js"></script>
 44   <script src="asset/js/bootstrap.min.js"></script>
 45 </body>
```

Offset

We can skip the columns using ‘offset’ as shown in Line 27; and the corresponding html page is shown in Fig. 3.4,

```

1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Bootstrap Tutorial</title>
5
6      <!-- CSS -->
7      <link href="asset/css/bootstrap.min.css" rel="stylesheet">
8      <!-- Add Custom CSS below -->
9      <style type="text/css">
10         .col-xs-1, .col-xs-2, .col-xs-3, .col-xs-4, .col-xs-5, .col-xs-6, .col-xs-7, .col-
11         xs-8,
12         .col-sm-1, .col-sm-2, .col-sm-3, .col-sm-4, .col-sm-5, .col-sm-6, .col-sm-7, .col-sm-
13         8,
14         .col-md-1, .col-md-2, .col-md-3, .col-md-4, .col-md-5, .col-md-6, .col-md-7, .col-md-
15         8,
16         .col-lg-1, .col-lg-2, .col-lg-3, .col-lg-4, .col-lg-5, .col-lg-6, .col-lg-7,
17         .col-lg-8 {
18             background-color: green;
19             color: white;
20             font-weight: bold;
21             border: 1px solid red;
22             height: 3em; /*height of the box*/
23             text-align: center; /*vertical center the text*/
24         }
25     </style>
26 </head>
27
28 <body>
29
30     <div class="row">
31         <div class="col-md-offset-2 col-md-2">col-md-2</div>
32         <div class="col-md-8">col-md-8</div>
33     </div>
34
35     <!-- Javascript -->
36     <!-- put jquery.js before bootstrap.min.js; and then add custom jquery -->
37     <script src="asset/js/jquery-3.3.1.min.js"></script>
38     <script src="asset/js/bootstrap.min.js"></script>
39 </body>
40 </html>

```

col-md-offset-2

col-md-2

col-md-8

Fig. 3.4: Offset

Components

Once we understand the 'html', 'css' and 'Bootstrap-grid' understood, then the use of Bootstrap-components are straightforward. Just copy and paste the code from the [Bootstrap website](#) and modify it according to the requirement.

In this section, we will use following template.

```

<!DOCTYPE html>
<html>
<head>
    <title>Bootstrap Tutorial</title>
    <!-- CSS -->
    <link href="asset/css/bootstrap.min.css" rel="stylesheet">
    <!-- Add Custom CSS below -->
</head>

```

(continues on next page)

(continued from previous page)

```

<body>

  <!-- J a v a s c r i p t -->
  <!-- put jQu e r y . j s b e f o r e b o o t s t r a p . m i n . j s ; a n d t h e n a d d
  c u s t o m j q u e r y -->
  <script src="a s s e t / j s / j q u e r y - 3 . 3 . 1 . m i n . j s"></script>
  <script src="a s s e t / j s / b o o t s t r a p . m i n . j s"></script>
</body>

```

Labels

- The class 'label' is used to create the label (Line 11).
- The class 'label-info' sets the color of the label to 'blue' (Line 11).
- The size of the 'labels' can be changed using 'element' tags (e.g. 'h3' in Line 11) outside the 'span' tag (Line 11).

Note: Bootstrap provides 6 color options,

- danger
- default
- info
- primary
- success
- warning

```

<!DOCTYPE html>
<html>
<head>
  4 <title>Bootstrap Tutorial</title>
  5 <!-- CSS -->
  6 <link href="a s s e t / c s s / b o o t s t r a p . m i n . c s s" r e l = " s t y l e s h e e t ">
  7 <!-- Add Custom CSS below -->
</head>
9
<body>
  <h3>Example heading <span class="label label-info">New</span></h3>
  12 <hr>
  13
  14 <!-- J a v a s c r i p t -->
  15 <!-- put jQu e r y . j s b e f o r e b o o t s t r a p . m i n . j s ; a n d t h e n a d d c u s t o m j q u e r y -->
  16 <script src="a s s e t / j s / j q u e r y - 3 . 3 . 1 . m i n . j s"></script>
  17 <script src="a s s e t / j s / b o o t s t r a p . m i n . j s"></script>
  18
</body>
</html>

```

- Fig. 3.5 is the output of above code,



Fig. 3.5: Label

Note: All the codes will be added in the 'body' tag, and the new code will be inserted below the previous codes. Therefore full HTML code is not added in the tutorial.

Buttons

The element 'button' can be used to create the button in Bootstrap as shown below,

Note:

- We can use 'lg', 'md' and 'xs' etc. to set the size of the button.
 - In Line 3, a glyphicon is added (i.e. trash sign). More glyphicon can be found on the [Bootstrap website](#).
-

```
<h3> Buttons </h3>
<button type="button" class="btn btn-primary">Sign in</button>
<button type="button" class="btn btn-danger">Delete <span class="glyphicon glyphicon-
trash"><span></
button>
<button type="submit" class="btn btn-lg btn-success">Submit</button>
<button type="submit" class="btn btn-md btn-success">Submit</button>
```

- Fig. 3.6 is the output of above code,



Fig. 3.6: Buttons

Forms

In this section, 4 types of forms are added,

3.4.3.1 Basic form

Output of below code is shown in Fig. 3.7,

```
<h2> Basic form </h2>
<div class="row">
<div class="col-sm-5" style="background: pink">
<div class="panel-body">
  <form role="form">
    <div class="form-group">
      <label for="exampleInputEmail">Email address</label>
      <input class="form-control" id="exampleInputEmail" placeholder="Enter email"
        type="email">
    </div>
    <div class="form-group">
      <label for="exampleInputPassword">Password</label>
      <input class="form-control" id="exampleInputPassword" placeholder="Password"
        type="password">
    </div>
  </form>
</div>
```

(continues on next page)

(continued from previous page)

```

<div class="form-group">
  <label for="exampleInputFile">File input</label>
  <input id="exampleInputFile" type="file">
  <p class="help-block">Example block-level help text here.</p>
</div>
<div class="checkbox">
  <label>
    <input type="checkbox"> Check me out
  </label>
</div>
<button type="submit" class="btn btn-info">Submit</button>
</form>
</div>
</div>
</div>
<br><br>

```

Basic form

Fig. 3.7: Basic form

Horizontal form

Output of below code is shown in [Fig. 3.8](#),

```

<h3> Horizontal form </h3>
<div class="row">

```

(continues on next page)

(continued from previous page)

```

<div class="col-sm-5" style="background: pink">
<div class="panel-body">
  <form class="form-horizontal" role="form">
    <div class="form-group">
      <label for="inputEmail1" class="col-lg-2 col-sm-2 control-label">Email</label>
      <div class="col-lg-10">
        <input class="form-control" id="inputEmail1" placeholder="Email"
          type="email">
        <p class="help-block">Example block-level help text here.</p>
      </div>
    </div>
    <div class="form-group">
      <label for="inputPassword1" class="col-lg-2 col-sm-2 control-label">Password</label>
      <div class="col-lg-10">
        <input class="form-control" id="inputPassword1" placeholder="Password"
          type="password">
      </div>
    </div>
    <div class="form-group">
      <div class="col-lg-offset-2 col-lg-10">
        <div class="checkbox">
          <label>
            <input type="checkbox"> Remember me
          </label>
        </div>
      </div>
    </div>
    <div class="form-group">
      <div class="col-lg-offset-2 col-lg-10">
        <button type="submit" class="btn btn-danger">Sign in</button>
      </div>
    </div>
  </form>
</div>

```

Inline form

Output of below code is shown in [Fig. 3.9](#),

```

<h2> Inline form </h2>
<div class="row">
<div class="col-sm-5" style="background: pink">
<div class="panel-body">
  <form class="form-inline" role="form">
    <div class="form-group">
      <label class="sr-only" for="exampleInputEmail2">Email address</label>
      <input class="form-control" id="exampleInputEmail2" placeholder="Enter email"
        type="email">
    </div>
    <div class="form-group">
      <label class="sr-only" for="exampleInputPassword2">Password</label>
      <input class="form-control" id="exampleInputPassword2" placeholder="Password"
        type="password">
    </div>
    <div class="checkbox">
      <label>
        <input type="checkbox"> Remember me
      </label>
    </div>
  </form>
</div>

```

(continues on next page)

Horizontal form

Fig. 3.8: Horizontal form

(continued from previous page)

```

</div>
</div>
</div>
<br><br>

```

Inline form

Fig. 3.9: Inline form

Modal form

Modal form appears in the pop-up window. Output of below code is shown in [Fig.](#)

```

<!-- modal form -->
<h3> Modal form </h3>

<div class="row">
<div class="col-sm-5" style="background: pink">

```


3.10,

(continues on next page)

(continued from previous page)

```

<div class="panel-body">
  <!-- button to generate model form -->
  <a href="#myModal" data-toggle="modal" class="btn
    btn-xs btn-success">Form in Modal
  </a>

  <!-- model form settings -->

  <div aria-hidden="true" aria-labelledby="myModalLabel" role="dialog" tabindex="-1"
    id="myModal"
    class="modal fade">
    <div class="modal-dialog">
      <div class="modal-content">
        <div class="modal-header">
          <button aria-hidden="true" data-dismiss="modal" class="close"
            type="button">&times;</
        <button>
          <h4 class="modal-title">Form Title </h4>
        </div>
        <div class="modal-body">

          <!-- actual form -->
          <form role="form">
            <div class="form-group">
              <label for="exampleInputEmail">Email address</label>
              <input class="form-control" id="exampleInputEmail3"
                placeholder="Enter email"
              type="email" </div>
              <div class="form-group">
                <label for="exampleInputPassword1">Password</label>
                <input class="form-control" id="exampleInputPassword3"
                  placeholder="Password"
                type="password" </div>
                <div class="form-group">
                  <label for="exampleInputFile">File input</label>
                  <input id="exampleInputFile3" type="file">
                  <p class="help-block">Example block-level help text here.</p>
                </div>
                <div class="checkbox">
                  <label>
                    <input type="checkbox"> Check me out
                  </label>
                </div>
                <button type="submit" class="btn btn-default">Submit</button>
              </form>
            <!-- actual form ends -->

          </div>
        </div>
      </div>
    </div>
  </div>
</div>
</div>
</div>
<hr>

```

Form elements

Output of below code is shown in [Fig. 3.11](#),

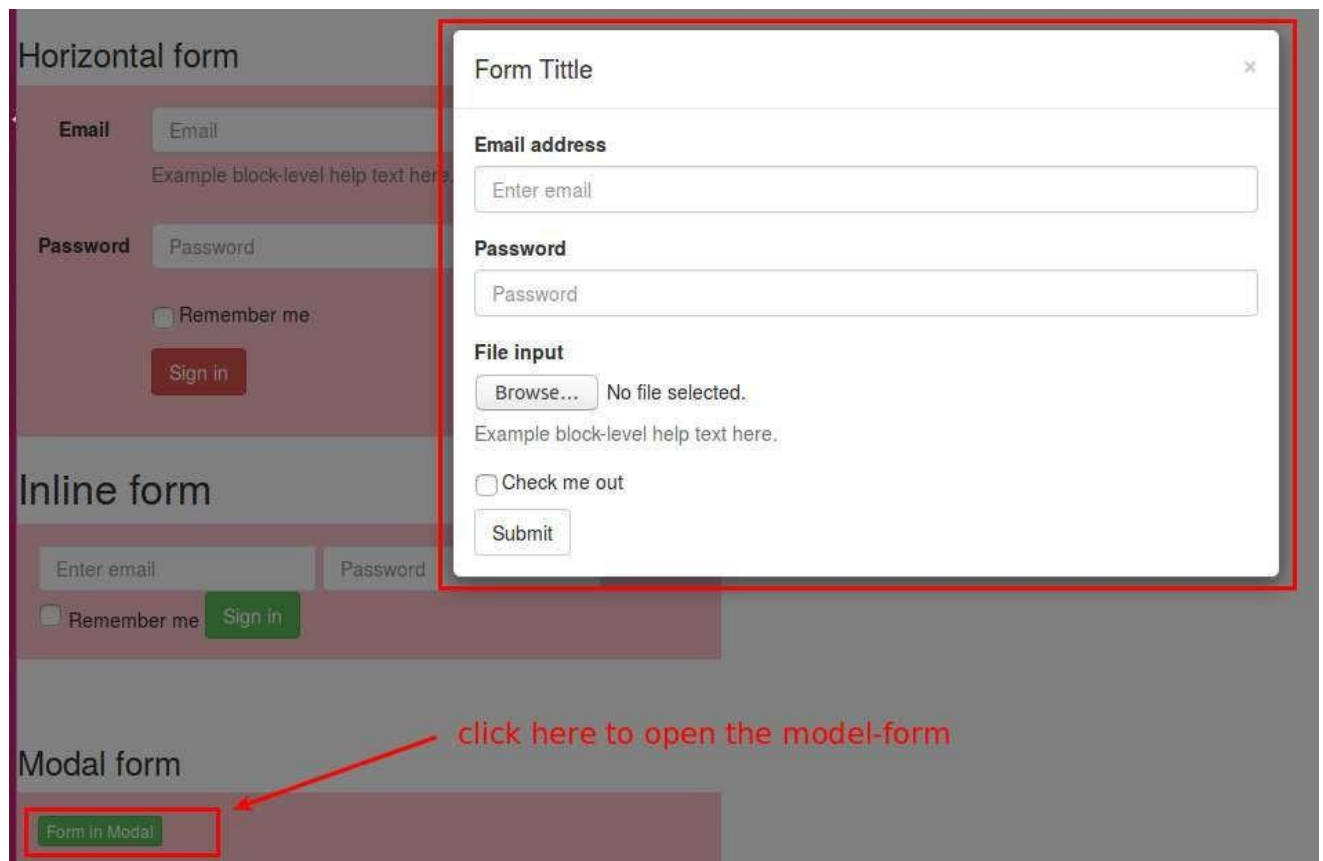


Fig. 3.10: Modal form

```

<!-- Form elements -->
<h3> Form elements </h3>
<div class="row">
  <div class="col-sm-5" style="background: pink">
    <div class="panel-body">
      <form class="form-horizontal" method="get">
        <div class="form-group">
          <label class="col-sm-2 col-sm-2 control-label">Default</label>
          <div class="col-sm-10">
            <input class="form-control" type="text">
          </div>
        </div>
        <div class="form-group">
          <label class="col-sm-2 col-sm-2 control-label">Help text</label>
          <div class="col-sm-10">
            <input class="form-control" type="text">
            <span class="help-block">A block of help text that breaks onto a new line and
            may
            extend beyond one line.</span>
          </div>
        </div>
        <div class="form-group">
          <label class="col-sm-2 col-sm-2 control-label">Rounder</label>
          <div class="col-sm-10">
            <input class="form-control round-input" type="text">
          </div>
        </div>
        <div class="form-group">
          <label class="col-sm-2 col-sm-2 control-label">Input focus</label>

```

(continues on next page)

(continued from previous page)

```

<div class="col-sm-10">
  <input class="form-control" id="focusedInput" value="This is focused..." type="text">
</div>
<div class="form-group">
  <label class="col-sm-2 col-sm-2 control-label">Disabled</label>
  <div class="col-sm-10">
    <input class="form-control" id="disabledInput" placeholder="Disabled input here..."
    disabled="disabled" type="text">
  </div>
</div>
<div class="form-group">
  <label class="col-sm-2 col-sm-2 control-label">Placeholder</label>
  <div class="col-sm-10">
    <input class="form-control" placeholder="placeholder" type="text">
  </div>
</div>
<div class="form-group">
  <label class="col-sm-2 col-sm-2 control-label">Password</label>
  <div class="col-sm-10">
    <input class="form-control" placeholder="" type="password">
  </div>
</div>
<div class="form-group">
  <label class="col-lg-2 col-sm-2 control-label">Static control</label>
  <div class="col-lg-10">
    <p class="form-control-static">email@example.com</p>
  </div>
</div>
</form>
</div>
</div>
</div>

```

Control size

Output of below code is shown in [Fig. 3.12](#),

```

<!-- Control size -->
<h3>Control size</h3>
<div class="row">
  <div class="col-sm-5" style="background: pink">
    <div class="panel-body">
      <form class="form-horizontal" method="get">
        <div class="form-group">
          <label class="col-sm-2 control-label col-lg-2" for="inputSuccess">Control sizing</label>
          <div class="col-lg-10">
            <input class="form-control input-lg m-bot15" placeholder=".input-lg" type="text">
            <input class="form-control m-bot15" placeholder="Default input" type="text">
            <input class="form-control input-sm m-bot15" placeholder=".input-sm" type="text">

            <select class="form-control input-lg m-bot15">
              <option selected="selected">Option 1</option>
              <option>Option 2</option>
              <option>Option 3</option>
            </select>
            <select class="form-control m-bot15">
              <option selected="selected">Option 1</option>

```

(continues on next page)

Form elements

Default	
Help text	 A block of help text that breaks onto a new line and may extend beyond one line.
Rounded	
Input focus	This is focused...
Disabled	Disabled input here...
Placeholder	placeholder
Password	
Static control	email@example.com

Fig. 3.11: Form elements

(continued from previous page)

```

        <option>Option 3</option>
      </select>
      <select class="form-control input-sm m-bot15">
        <option selected="selected">Option 1</option>
        <option>Option 2</option>
        <option>Option 3</option>
      </select>
    </div>
  </div>
</form>
</div>
</div>
</div>

```

Control size

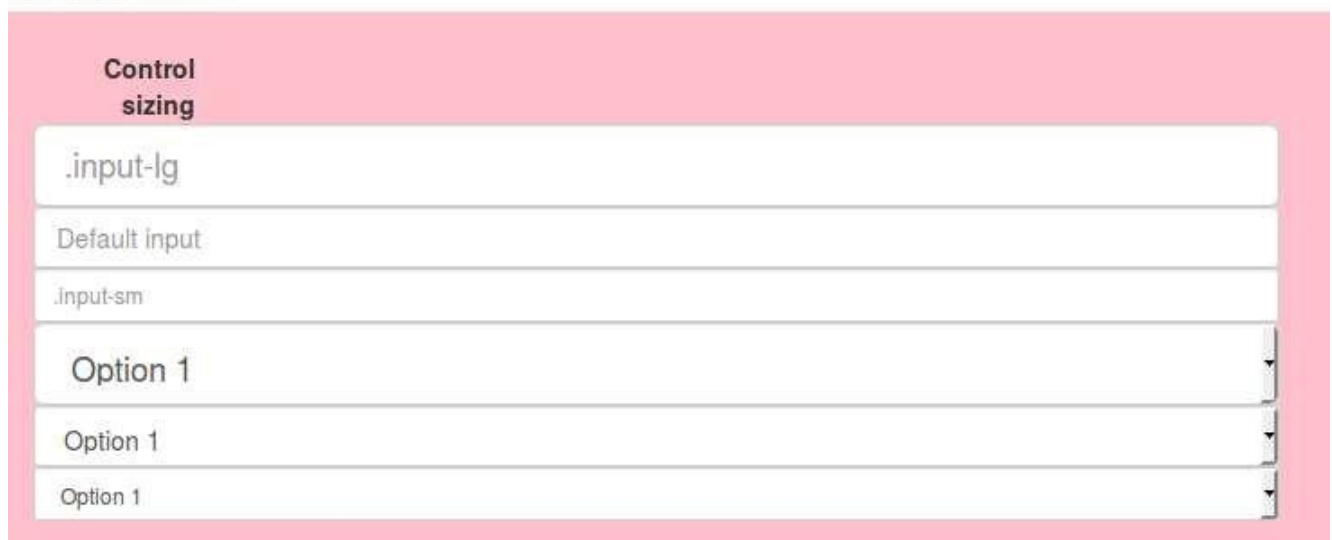


Fig. 3.12: Form elements

More buttons

Output of below code is shown in [Fig. 3.13](#),

```

<h3>Buttons</h3>
<div class="row">
  <div class="col-sm-12" style="background: pink">
    <div class="panel-body">
      <form class="form-horizontal" method="get">
        <div class="form-group">
          <label class="col-sm-2 control-label col-lg-2" for="inputSuccess">Checkboxes
            and
            radios</label>
          <div class="col-lg-10">
            <div class="checkbox">
              <label>
                <input value="" type="checkbox">
                  Option one is this and that. Be sure to include why it's great
                </label>
              </div>
            <div class="checkbox">

```

>

(continues on next
page)

(continued from previous page)

```

<label>
  <input value="" type="checkbox">
  Option one is this and that Be sure to include why it's great
  option one
</label>
</div>

<div class="radio">
  <label>
    <input name="optionsRadios" id="optionsRadios1"
    value="option1" checked=
    "checked"
    type="radio">
    Option one is this and that Be sure to include why it's great
  </label>
</div>
<div class="radio">
  <label>
    <input name="optionsRadios" id="optionsRadios2"
    value="option2" type=
    "radio">
    Option two can be something else and selecting it will deselect option_
    one
  </label>
</div>
</div>
<div class="form-group">
  <label class="col-sm-2 control-label col-lg-2" for="inputSuccess">Inline
  checkboxes</
  <div class="col-lg-10">
    <label class="checkbox-inline">
      <input id="inlineCheckbox1" value="option1" type="checkbox"> 1
    </label>
    <label class="checkbox-inline">
      <input id="inlineCheckbox2" value="option2" type="checkbox"> 2
    </label>
    <label class="checkbox-inline">
      <input id="inlineCheckbox3" value="option3" type="checkbox"> 3
    </label>
  </div>
</div>
<div class="form-group">
  <label class="col-sm-2 control-label col-lg-2"
  for="inputSuccess">Selects</label>
  <div class="col-lg-10">
    <select id="select-dropdown-num" class="form-control m-bot15">
      <option selected="selected">1</option>
      <option>2</option>
      <option>3</option>
      <option>4</option>
      <option>5</option>
    </select>

    <select multiple="multiple" class="form-control">
      <option>1</option>
      <option>2</option>
      <option>3</option>
      <option>4</option>

```

```
<      ion>5</option>
o    </select>
p </div>
</div>
```

(continues on next page)

(continued from previous page)

```

<div class="form-group">
  <label class="col-sm-2 control-label col-lg-2" for="inputSuccess">Column
  <div class="col-lg-10">
    <div class="row">
      <div class="col-lg-2">
        <input class="form-control" placeholder=".col-lg-2"
          type="text">
      </div>
      <div class="col-lg-3">
        <input class="form-control" placeholder=".col-lg-3"
          type="text">
      </div>
      <div class="col-lg-4">
        <input class="form-control" placeholder=".col-lg-4"
          type="text">
      </div>
    </div>
  </div>
</div>
</form>

```

Buttons

Checkboxes and radios

- ☐ Option one is this and thatâ€”be sure to include why it's great
- ☐ Option one is this and thatâ€”be sure to include why it's great option one
- ☒ Option one is this and thatâ€”be sure to include why it's great
- ☐ Option two can be something else and selecting it will deselect option one

Inline checkboxes

☐ 1 ☐ 2 ☐ 3

Selects

1

1
2
3
4

Column sizing

Fig. 3.13: Buttons

Input group

Output of below code is shown in [Fig. 3.14](#),

```

<!-- Input groups -->
<h3>Input groups</h3>
<div class="row">
<div class="col-sm-12" style="background: pink">
<div class="panel-body">
  <form class="form-horizontal" method="get">
    <div class="form-group">
      <label class="col-sm-2 control-label col-lg-2">Basic examples</label>
      <div class="col-lg-10">
        <div class="input-group m-bot15">
          <span class="input-group-addon">@</span>
          <input class="form-control" placeholder="User name" type="text">
        </div>

```

(continues on next page)

(continued from previous page)

```

        <div class="input-group m-bot15">
            <input class="form-control" type="text">
            <span class="input-group-addon">.00</span>
        </div>

        <div class="input-group m-bot15">
            <span class="input-group-addon">$</span>
            <input class="form-control" type="text">
            <span class="input-group-addon">.00</span>
        </div>
    </div>
</div>
<div class="form-group">
    <label class="col-sm-2 control-label col-lg-2">Sizing</label>
    <div class="col-lg-10">
        <div class="input-group input-group-lg m-bot15">
            <span class="input-group-addon">@</span>
            <input class="form-control input-lg" placeholder="Username" type="text">
        </div>

        <div class="input-group m-bot15">
            <span class="input-group-addon">@</span>
            <input class="form-control" placeholder="Username" type="text">
        </div>

        <div class="input-group input-group-sm m-bot15">
            <span class="input-group-addon">@</span>
            <input class="form-control" placeholder="Username" type="text">
        </div>
    </div>
</div>
<div class="form-group">
    <label class="col-sm-2 control-label col-lg-2">Checkbox and
    radio</label>
    <div class="col-lg-10">
        <div class="input-group m-bot15">
            <span class="input-group-addon">
                <input type="checkbox">
            </span>
            <input class="form-control" type="text">
        </div>

        <div class="input-group m-bot15">
            <span class="input-group-addon">
                <input type="radio">
            </span>
            <input class="form-control" type="text">
        </div>
    </div>
</div>
</div>
<div class="form-group">
    <label class="col-sm-2 control-label col-lg-2">Button addons</label>
    <div class="col-lg-10">
        <div class="input-group m-bot15">
            <span class="input-group-btn">
                <button class="btn btn-white" type="button">Go!</button>
            </span>
            <input class="form-control" type="text">
        </div>
    </div>
</div>

```

</div>

(continues on next page)

(continued from previous page)

```

<div class="input-group m-bot15">
  <input class="form-control" type="text">
  <span class="input-group-btn">
    <button class="btn btn-white" type="button">Go!</button>
  </span>
</div>

<div class="input-group m-bot15">
  <div class="input-group-btn">
    <button type="button" class="btn btn-white dropdown-toggle" data-
      toggle="dropdown"
→ ">Action <span class="caret"></span></button>
    <ul class="dropdown-menu">
      <li><a href="#">Action</a></li>
      <li><a href="#">Another action</a></li>
      <li><a href="#">Something else here</a></li>
      <li class="divider"></li>
      <li><a href="#">Separated link</a></li>
    </ul>
  </div>
  <input class="form-control" type="text">
</div>
<div class="input-group m-bot15">
  <input class="form-control" type="text">
  <div class="input-group-btn">
    <button type="button" class="btn btn-white dropdown-toggle" data-
      toggle="dropdown"
→ ">Action <span class="caret"></span></button>
    <ul class="dropdown-menu pull-right">
      <li><a href="#">Action</a></li>
      <li><a href="#">Another action</a></li>
      <li><a href="#">Something else here</a></li>
      <li class="divider"></li>
      <li><a href="#">Separated link</a></li>
    </ul>
  </div>
</div>
</div>

<div class="form-group">
  <label class="col-sm-2 control-label col-lg-2">Segmented buttons</label>
  <div class="col-lg-10">
    <div class="input-group m-bot15">
      <div class="input-group-btn">
        <button tabindex="-1" class="btn btn-white"
          type="button">Action</button>
        <button tabindex="-1" data-toggle="dropdown" class="btn btn-white
          dropdown-toggle
→
          type="button"
          n">
          <span class="caret"></span>
        </button>
        <ul role="menu" class="dropdown-menu">
          <li><a href="#">Action</a></li>
          <li><a href="#">Another action</a></li>
          <li><a href="#">Something else here</a></li>
          <li class="divider"></li>
          <li><a href="#">Separated link</a></li>
        </ul>
      </div>
    </div>
  </div>

```


</div>

<input class="form-control" type="text">

</div>

(continues on next page)

(continued from previous page)

```

<div class="input-group m-b-0 t15">
  <input class="form-control"
    type="text">
    <div class="input-group-btn">
      <button type="button" class="btn btn-white"
        type="button">Action</button>
      <button type="button" class="btn btn-white"
        type="button" data-toggle="dropdown">
        dropdown-toggle
        <span class="caret"></span>
      </button>
      <ul role="menu" class="dropdown-menu pull-right">
        <li><a href="#">Action</a></li>
        <li><a href="#">Another action</a></li>
        <li><a href="#">Something else here</a></li>
        <li class="divider"></li>
        <li><a href="#">Separated link</a></li>
      </ul>
    </div>
  </div>
</div>
</div>
</div>

```

Input groups

Fig. 3.14: Input groups

Navigation bar (navbar)

For navbar, one more 'css' file is added at Line 8. This CSS will create the space at the top for the 'navbar' so that it will not overlap with the contents of the HTML page. Further, Lines 13-44 adds various links to the navbar. Lastly, we can add HTML content below Line 46.

Output of below code is shown in Fig. 3.15,

```
<!DOCTYPE html>  
<html>  
<head>
```

(continues on next page)

(continued from previous page)

```

4     <title>Bootstrap Tutorial</title>
5     <!-- CSS -->
6     <link href="asset/css/bootstrap.min.css" rel="stylesheet">
7     <!-- Add Custom CSS below -->
8     <link href="asset/css/theme.css" rel="stylesheet"> <!-- make space at the top
for nav-bar -->
9 </head>
10
11 <body>
12
13     <div class="navbar navbar-inverse navbar-fixed-top">
14         <div class="container">
15             <div class="navbar-header">
16                 <button type="button" class="navbar-toggle" data-toggle="collapse"
data-target=".navbar-
collapse">
17                     <span class="icon-bar"></span>
18                     <span class="icon-bar"></span>
19                     <span class="icon-bar"></span>
20                 </button>
21                 <a class="navbar-brand hidden-sm"
href="http://pythondsp.readthedocs.io/">PythonDSP</a>
22             </div>
23             <div class="row">
24                 <div class="navbar-collapse collapse">
25                     <ul class="nav navbar-nav">
26
27                         <li><a href="basic.html">HTML</a></li>
28                         <li><a href="css.html">CSS</a></li>
29
30
31                         <li class="dropdown">
32                             <a href="#" class="dropdown-toggle" data-toggle="dropdown">More<b
class="caret"></b></a>
33                             <ul class="dropdown-menu">
34                                 <li><a href="bootstrap.html">Bootstrap</a></li>
35                                 <li><a href="js.html">JavaScript</a></li>
36                                 <li><a href="jquery.html">jQuery</a></li>
37                             </ul>
38                         </li>
39
40                     </ul>
41                 </div>
42             </div><!-- /nav-collapse -->
43         </div>
44     </div>
45
46     <!-- other codes here -->
47
48     <!-- JavaScript -->
49     <!-- put jQuery.js before bootstrap.min.js; and then add custom jquery -->
50     <script src="asset/js/jquery-3.3.1.min.js"></script>
51     <script src="asset/js/bootstrap.min.js"></script>
52 </body>
53 </html>

```

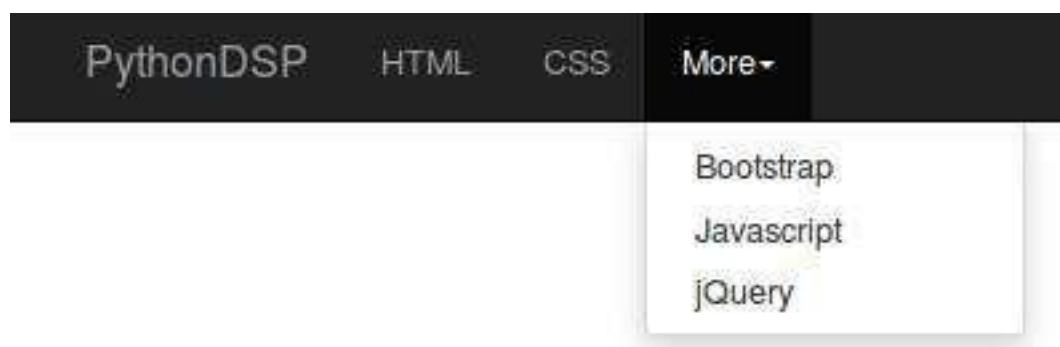


Fig. 3.15: N a v i g a t i o n b a r (n a v b a r)

Chapter 4

JavaScript

Introduction

- JavaScript is a dynamic language which is used for designing the web pages on the client side.
- It is case sensitive language.
- It is untyped language i.e. a variable can hold any type of value.
- // is used for comments.
- ; is used for line termination.
- JavaScript code should be added at the end i.e. **just above the closing-body-tag**.
- It is better to write the JavaScript code in separate file as shown in next section.

First code

The JavaScript code can be written in the 'html' file or in the separate 'JavaScript file (.js)' as shown in this section,

JavaScript in HTML file

In HTML file, the JavaScript code can be written inside the 'script' tag as shown in Lines 11-13 of below code. The code will write the message 'Hello World from JavaScript!' on the web page. Open the 'js.html' in the browser to see the message.

```
1 <!-- js.html -->
2
3 <!DOCTYPE html>
4 <html>
5 <head>
6   <title>JavaScript</title>
7 </head>
8
9
10
11 <script type="text/javascript">
12   document.write("Hello World from
13   JavaScript!<br>");
14 </script>
15
16 </body>
```

· Below is the output on the HTML page,

```

Hello World from JavaScript!

```

JavaScript in separate file

- The JavaScript code is saved in the file 'my_javascript.js' which is located inside the folders 'asset/js'. Note that, no 'script tag' is used here.

```

1 // asset/js/my_javascript.js
2 document.write("Hello World from JavaScript!<br>");

```

- Next we need to import the .js file in the HTML file as shown below,

```

1 <!-- js.html -->
2
3 <!DOCTYPE html>
4 <html>
5 <head>
6   <title>JavaScript</title>
7 </head>
8 <body>
9
10
11
12 <!-- import JavaScript files here -->
13
14 <script
15   src="asset/js/my_javascript.js"></script>
16
17 </body>

```

- Now, open the 'js.html' file in the browser and it will display the message.

Note: We will use the second method in this tutorial.

Keywords, Datatypes, Variables and Operators

Keywords

- Below are the reserved keywords in the JavaScript which cannot be used as 'variable' and 'function' names etc.

Table 4.1: List of keywords

Keywords			
abstract	boolean		break
	byte		
case	catch	char	class
const	continue	debugger	
	default	delete	do
	double	else	enum
	export	extends	false
final	float	for	
	function	goto	if
	implements		import
in	int	interface	long
native	new	null	
	package	private	
	protected		public
	return	short	static

Datatypes

JavaScript has three types of data,

- **Numbers** : 123, 32.32
- **Strings** : 'Meher' , 'Krishna Patel' , "123"
- **Boolean** : true, false

Note: All the numbers are considered as 'floating point' .

Variables

Variables can be defined using keyword 'var' . Further, JavaScript is untyped language i.e. a variable can hold any type of value .

- In the below HTML, 'p' tag with id 'p_name' is defined at Line 10. This id will be used to write some text using JavaScript,

```

<!-- js.html -->
2
<!DOCTYPE html>
<html>
<head>
6 <title>JavaScript</title>
</head>
8 <body>
9
11 <p id="p_name"></p>
12
13 <!-- import JavaScript files here -->
14 <script src="asset/js/my_javascript.js"></script>
15
</body>
16
</html>

```

- Two **variables** are defined at Lines 9-10 of type 'string' and 'float' respectively. Then **getElementById** is used to locate the tag with id 'p_name' and the text is inserted as shown in Line 11. Lastly, the '+' sign is used to concatenate the values as Line 11.

```
// 1
// 2
// 3
// print message on the webpage
document.write("Hello World from
6
7

// variable example
var your_name =
"Meher"; var age = 20;
document.getElementById("p_name").innerHTML = "Hello " + your_name +
```

- Below is the output of above codes. Note that the message 'Hello World from JavaScript!' is added at the end as 'JavaScript' file is imported at the end.

Warning: If we import the 'JavaScript' file between the 'ending-head-tag' and 'start-body-tag', then Message 'Hello Meher...' will not be displayed as 'JavaScript' will execute before the loading of the page; and JavaScript can not find the id 'p_name'.

```
Hello
Meher
Age : 20
```

Note: All the JavaScript/HTML codes will be added below the existing codes, therefore only the newly added code will be shown in the rest of the tutorial.

- 'prompt' can be used to read the values from the user.

```
// prompt
var x = prompt("enter a number");
document.write("2 * ", x, " = ", 2*x +
2*3 = 6
```

A pop-up window will appear due to prompt. If we provide the input value as 3, then below output will be generated,

Note: The ';' and '+' can be used to combine various outputs as shown in above example

Operators

Various operators are shown in this section. The usage of some of these operators are shown in [Section 4.4](#).

Arithmetic operators

- +
-
- *
- /
- % : modulus i.e remainder of the division
- ++ (increment)

• - (d e c r e m e n t)

Assignment operators

- =
- +=
- -=
- *=
- /=
- %=

Comparison operators

- == (compare only values not type)
- === (compare both values and type)
- !=
- >
- <
- >=
- <=

Conditional operator

- ?:

e.g. `( (a > b) ? a/b : b/a )` i.e if `a>b` then do `a/b` else do `b/a`

Logical operators

- && (logical and)
- || (logical or)
- ! (logical not)

Bitwise operators

- & (and)
- | (or)
- ^ (xor)
- ~ (not)

String to number conversion

'Number' is used to convert the string into numbers.

```
// string to number conversion
document.write("2 + Number("3.4") = ", 2 + Number("3.4"), "<br>");
2 + Number("3.4") = 5.4
```

Below is the output of above code,

Convert to integer

A string or float value can be converted into integer using 'parseInt'

```
// i n t c o n v e r s i o n
document.write("2 + parseInt('3.4') = ", 2 + parseInt("3.4"), "<br>"); // string to int
document.write("2 + parseInt(3.4) = ", 2 + parseInt(3.4), "<br>"); // float to int
```

Below are the outputs of above code,

```
2 + parseInt('3.4') = 5
2 + parseInt(3.4) = 5
```

Convert to float

The 'parseFloat' or 'Number' can be used to convert the value into float

```
document.write("2 + parseFloat('3.4') = ", 2 + parseFloat("3.4"), "<br>"); // parseFloat
values.
```

Math

'Math' library contains various useful functions as shown below,

```
// m a t h
document.write("pi = ", Math.PI,
pi = 3.141592653589793
<br>); // document.write("e = ",
e = 2.718281828459045
Math.E <br>);
similarly we can use 'abs', 'floor', 'ceil' and
document.write("Similarly, we can use 'abs', 'floor', 'ceil' and 'round' etc. <br>")
'round' etc. random number: 16
```

Below are the outputs of above code,

String

Below are the some of the useful 'string-styles' ,

```
// s t r i n g
document.write("m e h e r".toUpperCase(), "<br>") // uppercase

w = "Krishna"
document.write(w.toLowerCase(), "<br>") //
lowercase document.write(w.small(), "<br>") //
small document.write(w.bold(), "<br>") // bold
document.write(w.strike(), "<br>") // strike
document.write(w.fontSize("5em"), "<br>") // strike
document.write(w.link("http://python.dsp.rea.dth.edocs.io"), "<br>") // link
document.write(w.fontcolor("red").fontSize("12em"), "<br>") // multiple
```

The outputs are shown in [Fig. 4.1](#)

Arrays

In JavaScript, the arrays can store different types of values as shown in below code,

MEHER
 krishna
 Krishna
Krishna
 Krishna
 Krishna
Krishna
 Krishna

Fig. 4.1: String styles

```
// arrays
arr = [15, 30,
"Me her"] for(a in
arr)
    document.write(arr[a], " ");
document.write("<br>");

document.write(arr.pop(), "<br>"); //remove last
element arr.push("Krishna"); // add element to
end document.write(arr.pop(), "<br>");
```

Below are the output of above code,

```
15 30
Me her
Me her
Krishna
```

Control structure, loops and functions

If-else

In the below code, three conditions are checked for the variable 'age'; and

```
// asse t/js/my_j a v a script

// if- else
age = 10;
if (age > 3 && age < 6){
    document.write("Age : " + age + "<b> go to kindergarten</b>");
}
else if ( age >=6 && age < 18){
    document.write("Age : " + age + "<b> go to school</b>");
}
else{
    document.write("Age : " + age + "<b> go to college</b>");
}
```

corresponding message is printed.

(continues on next page)

(continued from previous page)

```
}
document.write("<br>");
```

- Since age is 10, therefore 'else if' statement is satisfied and we will get

```
Age : 10 go to school
```

below output,

Switch-case-default

Below is an example of Switch-case-default statement,

```
// a s s e t / j s / m y _ j a v a s c r i p t
Grade A : Very good grade!
// switch-case
var grade = 'A';
document.write("Grade " + grade
+ ": ");
switch(grade){
    case 'A':
        document.write("Very good
        grade!"); break;
    case 'B':
        document.write("Good
        grade!"); break;
    default: // if grade is neither 'A' nor 'B'
        document.write("Enter correct grade");
}
document.write("<br>");
```

- Below is the output of above code,

For loop

Below code prints the value from 5-0,

```
// For loop
for (i=5; i>=0; i--){
5 4 3 2 1 0
document.write(i + "
");
}
```

- Below is the output,

While loop

Below code prints the value from 0-4,

```
// While loop
x=0;
while (x < 5){
    document.write(x + "
"); x++;
}
document.write("<br>");
```


do-while

Below code prints the value from 0-2,

```
// do-while
x=0;
do{
    document.write(x + "
"); x++;
}while(x < 3);
document.write("<br
");
```

Below is the output,

for-in loop

The 'for-in' loop can be used to iterate over the array as shown below,

```
// for-in loop
arr = [10, 12, 31]; // array
for(a in arr){
    document.write(arr[a]
+ " ");
}
```

Continue and break

Continue and break statements are used inside the 'loop' statements.

- Continue will skip the loop and go to next iteration, if the condition is met. In the below code, 3 will not be printed at the output.

```
// continue
for(i=5; i>=0; i--){
5 4 2 if(i==3){ // skip 3
    continue;
}
    document.write(i + " ");
}
document.write("<br>");
```

Below is the output where 3 is not printed in the output.

- 'Break' statement will quit the loop if the condition is met. In the below code, the for loop will be terminated at 'i=3', therefore only 5 and 4

```
// break
for(i=5; i>=0; i--){
    if(i==3){ // exit loop when i=3
        break;
    }
    document.write(i + " ");
}
document.write("<br>");
```

will be printed,

Below is the output of the above code,

54

Functions

In the below code a function 'add2Num' is defined which adds the two numbers and

```
// function
function add2Num(num1, num2){ // function definition
    return num1 + num2;
}
sum = add2Num(2, 3); // function call
document.write("2 + 3 = " + sum);
document.write("<br>");
```

returns the result.

Below is the output,

2 + 3 = 5

Event handling

One of the main usage of JavaScript is 'event handling'. 'Mouse click' and 'pressing keys on keyboard' are the example of 'events'. With the help of JavaScript, we can perform certain actions at the occurrence of the events, as shown below.

- 'alert' is used to display message in the 'popup' window.

```
function alertMessage(message){
    alert(message)
}
```

- In Line 13, message 'Hello' is shown in the pop-up window on the event 'onclick'.
Note that **"JavaScript:void(0)"** is used here, which **does not** refresh the page.
- Line 15 is same as Line 13, but **"JavaScript:void(0)"** is not used, therefore page will be **refreshed**.
- Line 17 calls the function 'alertMessage' to display the message, which is defined in the above code.
- Lines 19-25 use different events i.e. 'onmouseover', 'onmouseleave', 'onmouseup' and 'onmousedown'; and the color and the text of the string changes based on these operations.

```
<!-- js.html -->
<!DOCTYPE html>
<html>
<head>
    <title>Java Script</title>
</head>
<body>
    <a href="JavaScript:void(0)", onclick="alert('Hello')">Say Hello</a><br> <!-- do not reload the page -->
    <a href="", onclick="alert('Hello')">Say Bye</a><br> <!-- reload the page -->
    <a href="JavaScript:void(0)", onclick="alertMessage('Bye via function')">Bye - function</a><br>
```

15
16
17
18
19

(continues on next page)

(continued from previous page)

```

    onmouseover="this.style.color='red'",
    onmouseleave="this.style.color='blue'", onmouseup="this.text='Not
    clicked'",
    onmousedown="this.text='You clicked'
26
27
28 <!-- import JavaScript files here -->
29 <script
30 src="asset/js/my_javascript.js"></script>
31
32 </body>

```

Conclusion

In this tutorial, we have selected the element with certain 'id' using 'getElementById'. Also, we saw the example of 'event handling'. Although, we can use JavaScript for these operations, but it is better to use **jQuery** for these purposes, which is discussed in [Chapter 5](#). A good approach is use to the jQuery for 'selectors' & 'event handling' and then use various features of JavaScript which are described in [Section 4.4](#) to implement the logics.

Chapter 5

jQuery

Introduction

jQuery is the JavaScript library.

Requirements

- First create a file 'jQuery.html'. Next, we need three files, i.e. one CSS (Line 9) and two JavaScript (Lines 23-24), for using the jQuery in this tutorial. These files are saved inside the folder 'asset', which can be downloaded from the repository. Lines 9 and 24 are the bootstrap files, whereas Line 23 is the actual 'jQuery' file which is required to use the jQuery.

Note: Note that, the 'jQuery.js' files should be placed above the 'bootstrap.js' file, otherwise the bootstrap-javascript will not work.

```
<!-- jQuery.html --
2
3
<!DOCTYPE html>
<html>
<head>
7 <title>jQuery</title>
8
  <!-- CSS -->
10 <link href="asset/css/bootstrap.min.css"
11 rel="stylesheet">
12 <!-- Add Custom styles below -->
13
14
15
16
17
18
19
20
21
22 <!-- JavaScript -->
23 <!-- put jQuery.js before bootstrap.min.js; and then add
  custom jQuery -->
  <script src="asset/js/jquery-3.3.1.min.js"></script>
26 <script src="asset/js/bootstrap.min.js"></script>
27
</body>
```


- If we open this html file in the browser, the a blank page will be displayed with header 'jQuery'.

Add contents

Next add one jumbotron (Line 15-19) and few labels (Lines 21-26) in the html file as shown in Fig. 5.1.

```

1 <!-- jQuery.html --
2 >
3
4 <!DOCTYPE html>
5 <html>
6 <head>
7   <title>jQuery</title>
8
9   <!-- CSS -->
10  <link href="asset/css/bootstrap.min.css" rel="stylesheet">
11  <!-- Add Custom styles below -->
12
13 </head>
14 <body>
15
16   <div class="jumbotron">
17     <div class="col-md-offset-4 container">
18       <h1><b>jQuery</b></h1>
19     </div>
20
21   </div>
22
23   <span class="label label-lg label-default"
24     id="I_default">Default</span>
25   <span class="label label-primary" id="I_primary">Primary</span>
26   <span class="label label-success" id="I_success">Success</span>
27   <span class="label label-info" id="I_info">Info</span>
28
29
30
31
32 <!-- JavaScript -->
33 <!-- put jQuery.js before bootstrap.min.js; and then add
34   custom jQuery -->
35 <script src="asset/js/jquery-3.3.1.min.js"></script>
36 <script src="asset/js/bootstrap.min.js"></script>
37
38 </body>

```

jQuery

Default Primary Success Info Warning Danger

Fig. 5.1: Jumbotron and labels are added

jQuery examples

Following are the important parts for writing a jQuery,

- **Selectors**: to select the specific element (e.g. 'id', 'class' etc.) for operations.
- **Event handling**: performs some operations (hide, toggle etc.) on the selected items at some events (e.g. click, mouse over etc.).

In this section, we will see the working of jQuery with the help of some examples. Then in [Section 5.3](#), we will see various methods to use the selectors to target the specific elements. Next, a list of operations are shown in [Table 5.2](#). Finally in [Section 5.5](#), a list of events are shown which can be used to initiate the operations.

Add jQuery code

Now, we can add the jQuery code as shown in Lines 36-40. Please note the following part,

- \$ sign is used to hit the jQuery. In the simple words, we can say that it execute the jQuery.
- The code is written between the following block. This block stops the jQuery to execute until the whole page is loaded, i.e. jQuery will be executed

```
$(function() {
    // write code here
});
```

after completion of the page-download.

- The above block is a shorthand for below,

```
$(document).ready(function() {
    // write code here
});
```

Note: We can use multiple blocks, if we want to separate the codes; or we can write all the jQueryes within one block.

- Lastly, the below line hides the id "l_default" after 1000 ms, i.e. 'Default' label will not be displayed after 1000 ms as shown in [Fig. 5.2](#)

```
$("#l_default").hide(1000); // hide id = 'l_default' using # operator after 300 ms
```

```
<!-- jQuery.html -->
2
<!DOCTYPE html>
<html>
<head>
6 <title>jQuery</title>
7
8 <!-- CSS -->
9 <link href="asset/css/bootstrap.min.css" rel="stylesheet">
10 <!-- Add Custom styles below -->
11
12 </head>
13 <body>
14
15 <div class="jumbotron">
16 <div class="col-md-offset-4 container">
17 <h1><b>jQuery</b></h1>
18 </div>
</div>
```

```
<span class="label label-lg label-default" id="l_default">Default</span>
```


19
20
21

(continues on next page)

(continued from previous page)

```

22 <span class="label label-primary"
23 id="l_primary">Primary</span>
24 <span class="label label-success"
25 id="l_success">Success</span>
26 <span class="label label-info" id="l_info">Info</span>
27
28
29
30
31 <!-- J a v a s c r i p t -->
32 <!-- put jquery.js before bootstrap.min.js; and then add custom jquery -->
33 <script src="asset/js/jquery-3.3.1.min.js"></script>
34 <script src="asset/js/bootstrap.min.js"></script>
35
36 <script type="text/javascript">
    $(function() {
        $("#l_default").hide(1000); // hide id = 'l_default' using # operator
        after 300 ms
    });
42 </script>
</body>

```



Fig. 5.2: Label 'Default' is hidden after 1000 ms

Note: '#' is used to select the 'id'; whereas '.' is used to access the 'class'.

- In the below code, the label 'Danger' is removed; and 'opacity= '0.5' ' is set for rest of the labels. Here, 'label' selects all the classes which have the name 'label' in it. Since all the entries have classes with name 'label', therefore all are selected and faded according to 'opacity= '0.5' ', as shown in Fig. 5.3.

```

<!-- jquery.html -->
2
<!DOCTYPE html>
<html>
<head>
6 <title>jQuery</title>
7
8 <!-- CSS -->
9 <link href="asset/css/bootstrap.min.css" rel="stylesheet">
10 <!-- Add Custom styles below -->
11
12 </head>
13 <body>
14
15 <div class="jumbotron">
16 <div class="col-md-offset-4 container">
17 <h1><b>jQuery</b></h1>
18 </div>
19 </div>
20
21
22 <span class="label label-lg label-default" id="l_default">Default</span>
23 <span class="label label-primary" id="l_primary">Primary</span>
    <span class="label label-success" id="l_success">Success</span>

```

(continues on next page)

(continued from previous page)

```

24 <span class="label label-info" id="l_info">Info</span>
25 <span class="label label-warning"
26 id="l_warning">Warning</span>
27
28
29
30
31 <!-- J a v a s c r i p t -->
32 <!-- put jQuery.js before bootstrap.min.js; and then add custom jQuery -->
33 <script src="asset/js/jquery-3.3.1.min.js"></script>
34 <script src="asset/js/bootstrap.min.js"></script>
35
36 <script type="text/javascript">
37     $(function() {
38         $("#l_default").hide(1000); // hide id = 'l_default' using # operator
39         // after 300 ms
40         $(".label-danger").hide(); // hide class = 'label' using .
41         // operator
42         $(".label").css({opacity:'0.5'}); // add opacity to all classes with
43         // name 'label'
44     });

```



Fig. 5.3: Faded labels

jQuery in separate file

It is better to write the jQuery in the separate files, therefore the codes are more manageable.

- For this, first write the code in the separate file 'my_jquery.js' as below,

Note: 'html' is used to change the content of the 'id' as shown in Line 5 below listing (see Fig. 5.4)

```

// asset/js/my_jquery.js
2
$(function() {
4     $("#l_default").hide(1000); // hide id = 'l_default' using # operator after 300 ms
5     $(".label-danger").html("Shark is dangerous"); // change content using 'html'
6 });

```

- Next, import this file to 'html' as shown in Line 35 of below code,

```

<!-- jQuery.html -->
2
<!DOCTYPE html>
<html>
<head>
6 <title>jQuery</title>
7
8 <!-- CSS -->
9 <link href="asset/css/bootstrap.min.css" rel="stylesheet">
10 <!-- Add Custom styles below -->
11

```

(continues on next page)

(continued from previous page)

```

12 </head>
13 <body>
14
15   <div class="jumbotron">
16     <div class="col-md-offset-4 container">
17       <h1><b>jQuery</b></h1>
18     </div>
19   </div>
20
21   <span class="label label-lg label-default" id="I_default">Default</span>
22   <span class="label label-primary" id="I_primary">Primary</span>
23   <span class="label label-success" id="I_success">Success</span>
24   <span class="label label-info" id="I_info">Info</span>
25   <span class="label label-warning" id="I_warning">Warning</span>
26   <span class="label label-danger" id="I_danger">Danger</span>
27
28
29
30
31   <!-- Javascript -->
32   <!-- put jquery.js before bootstrap.min.js; and then add custom jquery -->
33   <script src="asset/js/jquery-3.3.1.min.js"></script>
34   <script src="asset/js/bootstrap.min.js"></script>
35   <script src="asset/js/my_jquery.js"></script>
36
37 </body>
38 </html>

```



Fig. 5.4: Changed content using 'html' option

Get input from user

Below is the complete HTML code, which we will use in this section,

Listing 5.1: complete HTML code

```

<!-- jQuery.html -->
2
<!DOCTYPE html>
<html>
<head>
6   <title>jQuery</title>
7
8   <!-- CSS -->
9   <link href="asset/css/bootstrap.min.css" rel="stylesheet">
10  <!-- Add Custom styles below -->
11
12 </head>
13 <body>
14
15   <div class="jumbotron">
16     <div class="col-md-offset-4 container">
17       <h1><b>jQuery</b></h1>
18     </div>
19   </div>

```

(continues on next page)

(continued from previous page)

```

20
21 <span class="label label-lg label-default" id="l_default">Default</span>
22 <span class="label label-primary" id="l_primary">Primary</span>
23 <span class="label label-success" id="l_success">Success</span>
24 <span class="label label-info" id="l_info">Info</span>
25 <span class="label label-warning" id="l_warning">Warning</span>
26 <span class="label label-danger" id="l_danger">Danger</span>
27
28 <br><br>
29 <span class="btn btn-sm btn-warning" id="b_warning">Toggle Warning Label</span>
30
31 <br><br>
32 <!-- label_id is custom attribute whose values are set according to id of
   above labels-->
33 <span class="btn btn-sm btn-default label-btn" label_id='l_default'>Toggle Default
   Label</span>
34 <span class="btn btn-sm btn-primary label-btn" label_id='l_primary'>Toggle Primary
   Label</span>
35 <span class="btn btn-sm btn-success label-btn" label_id='l_success'>Toggle Success
   Label</span>
36 <span class="btn btn-sm btn-info label-btn" label_id='l_info'>Toggle Info
   Label</span>
37 <span class="btn btn-sm btn-warning label-btn" label_id='l_warning'>Toggle Warning
   Label</span>
38 <span class="btn btn-sm btn-danger label-btn" label_id='l_danger'>Toggle Danger
   Label</span>
39
40
41 <!-- J a v a s c r i p t -->
42 <!-- put jQuery.js before bootstrap.min.js; and then add custom jquery-->
43 <script src="asset/js/jquery-3.3.1.min.js"></script>
44 <script src="asset/js/bootstrap.min.js"></script>
45 <script src="asset/js/my_jquery.js"></script>
46
47 </body>
48 </html>

```

Toggle label on mouse click

In this section, the 'mouse click' on the button 'Toggle Warning Label' (Line 29) is read; and then the label 'warning' is toggled.

- The method 'on' waits for user's input, 'click' (i.e. mouse click).
- On mouse-click next query i.e. 'toggle' is executed which hides/unhides the 'warning label' as shown in Fig. 5.5 - Fig. 5.6.

```

// asset/js/my_jquery.js
2
// 3'hide label' and 'change content'
$(function() {
5   $("#l_default").hide(1000); // hide id = 'l_default' using # operator
6   after 300 ms
7   $(".label-danger").html("Shark is dangerous"); // change content using
8
9
// toggle 'warning label' based on click on the button "Toggle
Warning Label"
$(function() {
    $("#b_warning").on('click', function() {
        $("#l_warning").toggle();
    });
});

```

Toggle label based on the buttons

In previous section, we wrote the code which can toggle only one label. In this section, we will toggle multiple labels by writing one-generalized-jQuery as shown in [Fig. 5.7](#).



Fig. 5.5: Hide label 'warning'



Fig. 5.6: Show label 'warning'

- Lines 21-26 and 31-38 of [Listing 5.1](#) are used for this section,
- Custom attributes `label_id=...` are added in each button which has the value which corresponds to one of the 'label' `id`. e.g. `label_id` for Line 33 (i.e. `l_default`) is same as `id` at Line 21.
- Next, in the below jQuery, we saved the `label_id` in a variable (see Line 21) and then the variable is used to toggle the labels.

Listing 5.2: Toggle the label based on the button-clicked

```
// 1 a s s e t / j s / m y _ j q u e r y . j s
2
3 // 'hide label' and 'change content'
$(function() {
5     $("#l_default").hide(1000); // hide id = 'l_default' using # operator
6     after 300 ms
7     $(".label-danger").html("Shark is dangerous"); // change content using
8
9
10 // toggle 'warning label' based on click on the button 'Toggle
    Warning Label'
$(function() {
13     $("#b_warning").on("click", function() {
14         $("#l_warning").toggle();
15     });
16
17
18 // toggle panels according to the clicked-buttons
$(function() {
    $(".label-btn").on("click", function() { // select class 'label-btn'
        var labelId = $(this).attr("label_id"); // store the attribute
```



Fig. 5.7: Toggle labels based on button-clicked

(continued from previous page)

```

    $('#'+labelId).toggle(); // toggle the label with id stored
  });
};

```

Selectors

In this section, we will use following HTML to understand the various combination of the selectors which are summarized in [Table 5.1](#),

```

<!DOCTYPE html>
<html>
<head>
  <title>jQuery Selectors</title>
</head>
<body>
  <p>jQuery Selectors</p>

  <p class="foo"> Paragraph with class.</p>
  <p><span>Paragraph with span.</span></p>

  <p id="bar">Paragraph with ID. <span class="foo">Span with class and inside the
  paragraph-with-id.</
  →span></p>

  <div>
    <p my-id="my_para">Paragraph (with custom id) inside the div</p>
    <p my-id="my_para2"><span>Span inside the paragraph (with custom id) inside the

  <!-- J a v a s c r i p t -->
  <script src="asset/js/jquery-
  3.3.1.min.js"></script>

</body>

```

Note:

- Press 'ctrl+shift+k' (or go to Tools->Web Developer->Web Console) to open the web-console in **Firefox** and click on 'console' and type the 'jQuery' commands there.
- Press 'ctrl+shift+J' (or go to Tools->JavaScript Console) in **Chrome** and type the jQuery commands there.
- In this tutorial, the output of the 'Firefox' are shown.

Select elements

Select HTML elements

HTML elements, e.g. div, p, span and table etc., can be selected by writing them with the quotation mark as shown in below codes.

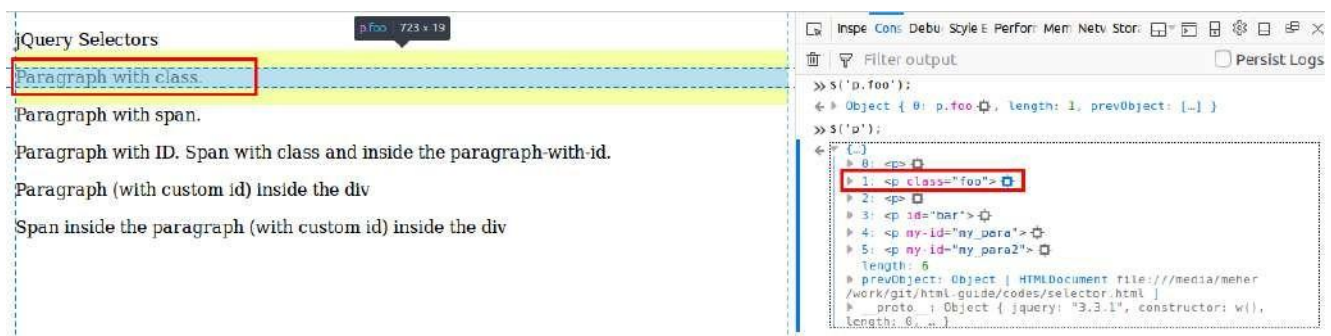
- Here, '\$('p')' is the command, whereas the lines below it are the outputs.
- The outputs show the 'selected elements' ; and the total number of selected elements i.e. 'length: 6' .
- If we put the cursor to one of the outputs, then it will highlight the

corresponding elements in the html as shown in [Fig. 5.8](#).

```

$('p')
{...}
0: <p>
1: <p class="foo">
2: <p>
3: <p id="bar">
4: <p my-id="my_para">
5: <p my-id="my_para2">
length: 6

```

Fig. 5.8: Output of `$( 'p' )`; (see red-squares)

- Similarly, we can select 'div' elements

```

$('div')
{...}
0: <div>
length: 1

```

Select class

Class can be selected using `'.'` operator. In below code, class `'foo'` is selected.

```

$(".foo")
{...}
0: <p class="foo">
1: <span class="foo">
length: 2

```

Select ID

ID can be selected using `'#'` operator. In the below code, id `'bar'` is selected.

```

$("#bar")
{...}
0: <p id="bar">
length: 1

```

Combining selectors

We can select an element with specific class name using `'.'` operator as shown in below code. Here element `'span'` with class `'foo'` is selected,

```
$("#span.foo")
{...}
0: <span class="foo">
length: 1
```

Multiple selectors

Use “,” to select different types of elements. In the below code, “paragraph with id ‘bar’” and “elements with class ‘foo’” are selected,

```
$("#p#bar, .foo")
{...}
0: <p class="foo">
1: <p id="bar">
2: <span class="foo">
length: 3
```

Select descendant element

Select ‘span’ inside the ‘p’ (use space between ‘span’ and ‘p’),

```
$("#p span")
{...}
0: <span>
1: <span class="foo">
2: <span>
length: 3
```

Select child element

The ‘>’ is used to select the child of an element.

```
$("#div>span")
{...}
length: 0
```

Note: There is one element inside ‘div’ but its path is ‘div->p->span’ therefore ‘span’ is a child of ‘p’ (not of ‘div’). But this ‘span’ is the descendant of ‘div’ therefore can be accessed using space,

```
$("#div span")
{...}
0: <span>
length: 1
```

Attribute selector

Custom attributes can be selected as below,

```
$("#[my-id = my_para]")
{...}
0: <p my-id="my_para">
length: 1
```

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(continued from previous page)

```

$('p[my-id != my_para]') // select p with 'my-id != my_para'
{...}
0: <p>
1: <p class="foo">
2: <p>
3: <p id="bar">
4: <p my-id="my_para2">
length: 5

$('p[class = foo]')
{...}
0: <p class="foo">
length: 1

```

Filters

Various filter options (e.g. filter, first, last and has etc.) can be used after ':' as

```

$('p:first') // select first
element
{...}

$('p:last') // last element
{...}
0: <p my-id="my_para2">
length: 1

$('p:not(foo)') // 'p' which does not have class 'foo'
{...}
0: <p>
1: <p>
2: <p id="bar">
3: <p my-id="my_para">
4: <p my-id="my_para2">
length: 5

$('p:has(span)') // 'p' which has 'span' inside it
{...}
0: <p>
1: <p id="bar">
2: <p my-id="my_para2">
length: 3

$('p:hidden') // hidden 'p'
{...}
length: 0

$('p:visible') // visible 'p'
{...}
0: <p>
1: <p class="foo">
2: <p>
3: <p id="bar">
4: <p my-id="my_para">
5: <p my-id="my_para2">
length: 6

$('p:contains(with class)') // paragraph which contains class

```

shown below,

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```

{...}
0: <p class="foo">
1: <p id="bar">
length: 2

$('p').filter(".foo") // same as p.foo
{...}
0: <p class="foo">
length: 1

$('p').not(".foo") // p with not class foo
{
.
.
.
}

0
:

<
p
>
1: <p>
2: <p id="bar">
3: <p my-id="my_para">
4: <p my-id="my_para2">
length: 5

$('p').last() // last element of 'p'
{...}
0: <p my-id="my_para2">
length: 1

$('p').has('span') // p with span
{
.
.
.
}

0
:

<
p
>
1: <p id="bar">
2: <p my-id="my_para2">
length: 3

$('p').is('.foo') // does 'p'
contain 'foo' true

$('p').children() // children of 'p'
{...}
0: <span>
1: <span class="foo">
2: <span>
length: 3

```



```

$('p').children(".foo") // child of 'p' with class 'foo'
{...}
0: <span class="foo">
length: 1

```

Table 5.1: Selectors

Command	Description	example
<code>\$('elementName')</code>	select the tags (p, body, table)	<code>\$('p'), \$('body')</code>
<code>\$('className')</code>	select element with 'className'	<code>\$(".foo")</code>
<code>\$('elementName.className')</code>		element with
<code>className</code>	<code>\$("span.foo")</code>	
<code>\$(selector1, selector2)</code>	select selector1 and selector2	<code>\$('p#bar, .foo')</code>
<code>\$(selector1 selector2)</code>	select selector2 inside selector1	<code>\$('p span')</code>
<code>\$(selector1>selector2)</code>	select selector2 which is child of selector1	
	<code>\$('div>span')</code>	
<code>\$(a ttributName = someName)</code>		select a ttribute
with someName	<code>\$('[my-id=my_para]')</code>	
<code>\$p(a ttributName = someName)</code>		select 'p'
with a ttribute with someName		<code>\$('p[my-id=</code>
<code>my_para]')</code>	See Section 5.3.2 for filters	

Operations

In Listing 5.2 we saw some operations e.g. 'hide()' and 'toggle()'. Table 5.2 shows some more such operations.

Table 5.2: Operations

Operations			
hide()	hide(speed)		
	fadeIn(speed)		
	fadeOut(speed)	show()	show(speed)

Event handling

In Listing 5.2 we saw one example of event handling, where certain operations are performed on the event 'mouse click'. Table 5.2 shows a list of some more events.

Table 5.3: Event handling

Events			
click	keydown	keypress	
mousedown	on		